

Workshop on Essentials for Mortality Investigation of Asian Elephant



5–6 June 2026

Date

Raigarh, CG

Venue

84 Officers

Participants

8 Modules

Chapters

Organised by: Chhattisgarh Forest Department

Technical Support: WII Dehradun · Lab Partners: ICAR-IVRI & NDVSU Jabalpur

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- : WORKSHOP OVERVIEW : -

Workshop on Essentials for Mortality Investigation of Asian Elephant

Chhattisgarh Forest Department · Bilaspur Circle

In Technical Partnership with Wildlife Institute of India & ICAR-IVRI

05 – 06 June 2026 · Raigarh, Chhattisgarh

2

Days

18

Sessions

5

National
Experts

30+

Participants

48

Elephant
Deaths Analyzed



Inaugural session — Workshop on Mortality Investigation of Asian Elephant, Raigarh, June 2026.

The **Workshop on Essentials for Mortality Investigation of Asian Elephant** was organized by the Chhattisgarh Forest Department, Bilaspur Circle, through the special initiative of **Hon'ble Forest Minister Shri Kedar Kashyap**, in technical partnership with the **Wildlife Institute of India**, Dehradun, and **ICAR-IVRI**, Bareilly. Held on 05–06 June 2026 at Raigarh, Chhattisgarh, the two-day workshop brought together five nationally eminent experts to equip field officers with systematic, evidence-based skills for elephant mortality investigation.



- : Organizing Team : -

Shri Kedar Kashyap

Hon'ble Forest Minister

Department of Forest & Climate Change, Government of Chhattisgarh

Shri Arun Kumar Pandey, IFS

Principal Chief Conservator of Forests & Head of Forest Force

Department of Forest & Climate Change, Government of Chhattisgarh

Senior Officers

Shri Matheshwaran V., IFS

APCCF Wildlife

CG Forest Dept.

Shri Manoj Kumar Pandey, IFS

CCF, Bilaspur Circle

CG Forest Dept.

Smt. Priyanka Pandey, IFS

CCF Wildlife Bilaspur
& Field Director ATR
CG Forest Dept.

Technical Partners

Wildlife Institute of India (WII), Dehradun — India's premier institution for wildlife research and training, contributing expertise in wildlife health management, chemical capture, and wildlife forensic science.

ICAR-Indian Veterinary Research Institute (IVRI), Bareilly — National centre for veterinary pathology, wildlife disease diagnosis, and health surveillance, contributing expertise in necropsy, molecular diagnostics, and sampling protocols.

Nanaji Deshmukh Veterinary Science University (NDVSU), Jabalpur — Contributing expertise in wildlife forensics and health management drawn from three decades of field experience across Central India.

Resource Persons

01 Dr. Parag Nigam

Session Chair

Senior Scientist & Head, Wildlife Health Management

Wildlife Institute of India, Dehradun

02 Dr. A. B. Shrivastav

Resource Person

Former Director, Centre for Wildlife Forensic & Health; Professor & Veterinary Pathologist

NDVSU, Jabalpur

03 Dr. Karikalan Mathesh

Resource Person

Senior Scientist, Centre for Wildlife Conservation, Management & Disease Surveillance

ICAR-IVRI, Bareilly

04 Dr. Chandra Prakash Sharma

Resource Person

Principal Technical Officer & In-charge, Morphology Unit, WFCGC

- : Speaker Statements : -

DA

"Every elephant death is a data point — if we investigate systematically, the forest tells us exactly what we need to know to prevent the next one."

– Dr. A. B. Shrivastav

DP

"Systematic necropsy is not just a scientific procedure; it is an act of accountability to the elephant, to the ecosystem, and to the communities that share this landscape."

– Dr. Parag Nigam

DK

"EEHV is a silent predator. Without PCR diagnostics reaching the ground level, we remain blind to the most lethal infectious threat young elephants face today."

– Dr. Karikalan Mathesh

DC

"Evidence collected poorly is evidence lost forever. Forensic discipline at the mortality site determines the outcome of every investigation — and every court proceeding."

– Dr. Chandra Prakash Sharma

DT

"Every tissue sample we preserve today is a resource for tomorrow's conservation science. The necropsy table is where field work meets molecular genetics."

– Dr. Tapendra Saini

■ ■ Workshop Background

Chhattisgarh has emerged as one of India's most significant elephant habitats, with the Bilaspur Circle — encompassing the Dharamjaigarh and Raigarh Van Mandals — hosting the state's highest concentration of wild elephants. The elephant population has grown remarkably, from an estimated 24 individuals in 2001 to approximately 450 elephants in 2026, reflecting nearly 10% compound annual growth. These elephants inhabit forest areas across the Surguja, Bilaspur, Raipur, and Durg divisions. This rapid expansion has intensified the human-elephant interface and, critically, raised the toll of elephant mortalities to a level demanding systematic institutional response.

Between 2021–22 and 2026–27, a total of 48 elephant deaths were recorded across Dharamjaigarh (27 deaths) and Raigarh (21 deaths) Van Mandals. Electrocution (51%) and drowning (22%) account for the majority. The 2025–26 season recorded the highest single-year toll, with 2026–27 already showing a disturbing pattern of calf drowning deaths in the early months of the year.

— Chhattisgarh Forest Department Data, 2021–2026

In view of the incidents of elephant calf deaths in the Raigarh and Dharamjaigarh Forest Divisions, and to understand the causes of such deaths, a national-level workshop was organized on 05–06 June 2026 in Raigarh district through the special initiative of **Hon'ble Forest Minister Shri Kedar Kashyap**. The Department of Forest & Climate Change is working to bring human-elephant conflict to zero, with positive results already witnessed in the field.

■ ■ Workshop Objectives

- To train field officers in systematic and scientific investigation of elephant mortalities, covering veterinary, forensic, and legal dimensions.
- To standardize necropsy procedures, sampling protocols, and mortality documentation practices for the Dharamjaigarh and Raigarh Van Mandals.
- To equip officers with the knowledge to identify major causes of elephant death — electrocution, drowning, infectious diseases, and poisoning — and implement targeted preventive measures.

■ ■ Participants

- **Chhattisgarh Forest Department:** Range Officers, Beat Guards, and Wildlife Wardens from Dharamjaigarh and Raigarh Van Mandals, and officers from elephant-affected districts.
- **Animal Resources Department, Chhattisgarh:** Veterinary officers from elephant-affected districts of the state.
- **National Technical Experts:** Veterinarians, scientists, and elephant specialists from Wildlife Institute of India, ICAR-IVRI, and NDVSU Jabalpur.

Inaugural Session

05 June 2026 · 10:00 – 11:00 hrs · Raigarh, Chhattisgarh

Shri Kedar Kashyap

Hon'ble Forest Minister — Inaugural Address

Department of Forest & Climate Change, Government of Chhattisgarh

The workshop was formally inaugurated by **Hon'ble Forest Minister Shri Kedar Kashyap**, whose special initiative drove the organization of this national-level event in response to the recent incidents of elephant calf deaths in the Raigarh and Dharamjaigarh Forest Divisions. In his inaugural address, the Hon'ble Minister spoke directly to the significance of this workshop — both as a conservation imperative and as a commitment to the communities living alongside Chhattisgarh's growing elephant population.

Full Address — Hon'ble Forest Minister Shri Kedar Kashyap

The elephant population in Chhattisgarh has been growing continuously over the past five years, and currently approximately 450 elephants are present in the state. These elephants are permanently inhabiting the forest areas of Surguja, Bilaspur, Raipur, and Durg divisions. The Department of Forest and Climate Change, Government of Chhattisgarh, is continuously working for the conservation, enhancement, and protection of these elephants while also ensuring the safety of local communities. Under the direction of the Hon'ble Forest Minister, the department is striving to bring human-elephant conflict to zero — and positive outcomes have already been achieved.

Keeping in mind the recent incidents of elephant calf deaths in the Raigarh and Dharamjaigarh Forest Divisions of Raigarh district, and in order to investigate the causes of these deaths, a national-level workshop has been organised on 5–6 June 2026 in Raigarh district through the special initiative of Hon'ble Forest Minister Shri Kedar Kashyap. This workshop brings together national-level veterinarians, scientists, and elephant experts alongside state veterinarians and officers of the Department of Forest & Climate Change and the Animal Resources Department from all elephant-affected districts of Chhattisgarh.

The objective of the workshop is to scientifically investigate the causes of wild elephant deaths and to prepare management strategies for the future. Participants will receive practical training in examining dead elephants, collecting biological samples, and dispatching them safely to laboratories. In addition, the workshop will strengthen safety protocols for carcass handling, procedures for proper carcass disposal, and overall preparedness for elephant mortality incidents and health surveillance systems across the state.

■ ■ Address by PCCF & Head of Forest Force

Shri Arun Kumar Pandey, IFS

Principal Chief Conservator of Forests & Head of Forest Force

Department of Forest & Climate Change, Government of Chhattisgarh

Shri Arun Kumar Pandey, IFS, Principal Chief Conservator of Forests and Head of Forest Force, Chhattisgarh, addressed the workshop with a comprehensive overview of the state's elephant conservation efforts and the critical institutional context that made this workshop necessary and timely.

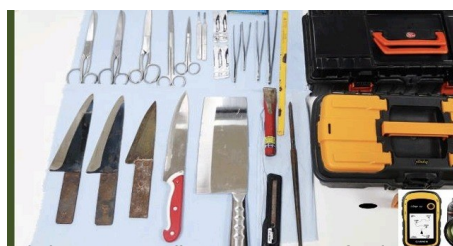
Shri Pandey outlined the remarkable growth of Chhattisgarh's elephant population — from 24 individuals in 2001 to approximately 450 today — as a testament to the department's sustained conservation work over two decades. He acknowledged, however, that this population growth also brought new and complex challenges: increasing human-elephant conflict, expanding corridor pressure, and, most urgently, a rising toll of elephant deaths that demanded systematic investigation.

He emphasized that the Chhattisgarh Forest Department, under the direction of Hon'ble Forest Minister Shri Kedar Kashyap, is committed to understanding the precise causes of every elephant death — particularly the recent incidents of calf mortality in the Raigarh and Dharamjaigarh Van Mandals. He expressed that bringing national-level expertise from WII, ICAR-IVRI, and NDVSU to the field — rather than sending samples to distant laboratories without trained local personnel — represented a fundamental shift in the department's approach to wildlife health management.

The strength of our forest department lies in its field officers. This workshop will give them the scientific tools to do their duty with the precision and accountability that these magnificent animals — and the communities living with them — deserve.

— Shri Arun Kumar Pandey, IFS, PCCF & HoFF

Shri Pandey called upon all participants to engage actively in both the theoretical and practical sessions, and to carry the learning back to their divisions with the commitment to establish standardized investigation protocols. He assured the resource persons of the department's full support in implementing the workshop's recommendations and developing a formal SOP for elephant mortality investigation in Chhattisgarh.



Inaugural ceremony — Workshop on Mortality Investigation of Asian Elephant, Raigarh, June 2026.

■ ■ Addresses by Senior Officers

Shri Matheshwaran V., IFS

Additional Principal Chief Conservator of Forests (Wildlife)

Department of Forest & Climate Change, Government of Chhattisgarh

Shri Matheshwaran V., APCCF Wildlife, addressed the gathering from the perspective of wildlife management across the state. He highlighted that the elephant mortality challenge in Chhattisgarh is inseparable from the broader question of wildlife corridor management and human-wildlife coexistence. He emphasized the critical role of systematic mortality investigation in building an evidence base that directly informs policy and management interventions. He expressed the Wildlife Wing's commitment to institutionalizing the protocols developed through this workshop and extending the training to other elephant-affected divisions across the state.

Shri Manoj Kumar Pandey, IFS

Chief Conservator of Forests, Bilaspur Circle

Department of Forest & Climate Change, Government of Chhattisgarh

As the hosting officer for the workshop, Shri Manoj Kumar Pandey, CCF Bilaspur Circle, welcomed all resource persons, participants, and senior officers. He placed the workshop in the immediate operational context of the Bilaspur Circle, where the Dharamjaigarh and Raigarh Van Mandals have recorded 48 elephant deaths over the past five years. He acknowledged that while the Circle's field officers are dedicated and experienced, the specialized knowledge required for systematic elephant mortality investigation — particularly in disease diagnosis, forensic documentation, and sample chain-of-custody — had long been a gap that this workshop was specifically designed to address. He expressed gratitude to the Hon'ble Minister and PCCF for their initiative and support in organizing this event.

Smt. Priyanka Pandey, IFS

CCF Wildlife, Bilaspur & Field Director, Achanakmar Tiger Reserve

Department of Forest & Climate Change, Government of Chhattisgarh

Smt. Priyanka Pandey, CCF Wildlife Bilaspur and Field Director of the Achanakmar Tiger Reserve, brought a unique perspective to the inaugural session, highlighting the intersection of elephant mortality investigation with tiger reserve management. She emphasized that the movement of elephants through the ATR landscape made robust mortality investigation protocols essential not just for elephant conservation, but also for the integrity of tiger reserve management. She welcomed the national faculty and expressed confidence that the two-day workshop would lay the foundation for a new standard of scientific rigour in wildlife mortality investigation across Chhattisgarh's protected area network.



Inaugural session with senior officers and resource persons, Raigarh, 05 June 2026.

■ ■ Inaugural Remarks by Resource Persons

Dr. Parag Nigam — Senior Scientist & Head, Wildlife Health Management, WII

Dr. Nigam, in his inaugural remarks, underscored the urgent need for evidence-based investigation of wildlife mortalities. Drawing from nearly three decades of field experience — including India's first tiger reintroduction at Sariska and landmark Indian Gaur reintroductions — he spoke to the critical importance of systematic field necropsy as the cornerstone of wildlife conservation medicine. He commended the Chhattisgarh Forest Department for its proactive approach and expressed that the WII–ICAR–IVRI partnership for this workshop represented a model of institutional collaboration that could be replicated across other states facing similar challenges.

Dr. A. B. Shrivastav — Former Director, Centre for Wildlife Forensic & Health, NDVSU

Prof. Shrivastav addressed the gathering with particular insight into the One Health dimension of elephant mortalities. He emphasized that diseases transmissible between elephants, livestock, and humans — particularly tuberculosis and anthrax — make systematic mortality investigation a public health imperative, not merely a wildlife conservation concern. He called for the workshop's outcomes to be converted into a formal State-level Standard Operating Procedure, backed by departmental order.

Dr. Karikalan Mathesh — Senior Scientist, ICAR-IVRI, Bareilly

Dr. Karikalan spoke on the diagnostic laboratory's perspective on field-collected samples. He emphasized that the quality of samples received from field mortality sites determines what can and cannot be diagnosed — and that poor sample collection is the single most common reason why causes of death remain undetermined. He expressed that equipping Chhattisgarh's field officers with proper sampling knowledge would directly improve the diagnostic yield from every future mortality case referred to ICAR-IVRI.

Dr. Chandra Prakash Sharma — Principal Technical Officer, WII

Dr. Sharma brought the forensic scientist's perspective to the inaugural session. He noted that in his decades of work supporting enforcement agencies across India, the most common weakness in wildlife mortality cases reaching courts was inadequate field documentation. He expressed confidence that the forensic and legal sessions of the workshop would address this gap directly, equipping officers to build cases that could survive legal scrutiny.

This workshop is not just about learning what to do — it is about learning what the law requires of you. Every forest officer at a mortality site is simultaneously a scientist, a custodian of evidence, and an officer of the law.

— Dr. Chandra Prakash Sharma, Wildlife Institute of India

■ ■ Workshop Programme

Two-Day Programme · 05–06 June 2026 · Raigarh, Chhattisgarh · 18 Sessions

Day 1 · 05 June 2026 · 10:00 – 17:30

10:00 11:00	Inauguration Official opening, address by Forest Minister, PCCF & HoFF, senior officers, and resource persons.	Forest Minister · PCCF · Senior Officers
11:00 11:30	Elephant Status & Human-Elephant Interface in Chhattisgarh Population dynamics, corridor use, conflict and mortality trends.	CG Forest Dept.
11:30 12:15	Biological & Anatomical Aspects of Elephants Key anatomical peculiarities relevant to field investigation.	Dr. P. Nigam
12:15 13:00	Understanding Wildlife Mortalities: The Veterinary Perspective Carcass condition, natural vs unnatural death; ante/post-mortem changes & time of death.	Dr. P. Nigam
— Lunch Break: 13:00 – 14:00 —		
14:00 15:30	Infectious Causes of Mortality in Elephants EEHV, EMC, FMD, Rabies, TB, Anthrax, Haemorrhagic Septicaemia.	Dr. K. Mathesh, Dr. Shrivastav
15:30 16:30	Non-Infectious & Disaster Mortalities Electrocution, lightning, train hits, infighting, natural disaster deaths.	Dr. Shrivastav, Dr. K. Mathesh
16:30 17:30	Non-Infectious Diseases in Elephants Mycotoxicosis, kodo millet poisoning, and environmental bio-toxins.	Dr. K. Mathesh, Dr. P. Nigam

Day 2 · 06 June 2026 · 09:30 – 17:30

09:30 10:15	Biosafety & Biosecurity at Elephant Mortality Sites Zoonotic risks, PPE requirements, safe carcass disposal, site sanitation.	Dr. K. Mathesh, Dr. Saini, Dr. Nigam
10:15 11:00	Equipment Familiarization & Site Preparation Chain saws, axes, winches; establishing a secure investigation perimeter.	Dr. Shrivastav, Dr. K. Mathesh, Dr. Saini
— Tea Break & Equipment Demonstration: 11:00 – 11:30 —		
11:30 12:30	Comprehensive Sampling Protocols Collection, packaging, cold chain, transport & documentation for lab analysis.	Dr. K. Mathesh, Dr. Shrivastav, Dr. Saini
12:30 13:00	Recording Information During Necropsy Report writing and photographic documentation standards.	Dr. K. Mathesh, Dr. Shrivastav
— Lunch Break: 13:00 – 14:00 —		
14:00 14:45	Vetro-Legal & Forensic Aspects of Mortality Investigation Investigation essentials for legal and forensic frameworks.	Dr. C.P. Sharma
14:45 16:00	Practical: Crime Scene Investigation Chain of custody, evidence collection, legal procedures for ivory/tusk extraction.	Dr. C.P. Sharma, Dr. Nigam, Dr. Saini
16:15 16:45	Panel Discussion Critical aspects of managing mortalities in elephants.	All Faculty & CG Officers
16:45 17:30	Valedictory Workshop recap, feedback collection, and closing ceremony.	—

■ ■ Session Highlights · Day 1 · 05 June 2026

● Elephant Status & Human-Elephant Interface in Chhattisgarh

The opening technical session presented a comprehensive overview of elephant population dynamics in the Bilaspur Circle. Data revealed a growth trajectory from 24 elephants in 2001 to approximately 450 by 2026, with the Dharamjaigarh and Raigarh Van Mandals as the primary habitat zone. Year-wise mortality data showing 48 deaths over five years was presented, with electrocution (51%) and drowning (22%) as dominant causes. The intensification of conflict as agricultural expansion encroaches on corridors was identified as the central management challenge.

● Biological & Anatomical Aspects of Elephants

Dr. Parag Nigam delivered an engaging session on the unique anatomy and physiology of Asian elephants relevant to field investigation — including foot structure, dental formula and tusk anatomy, organ positions, and why elephant necropsy requires specialized equipment and a coordinated multi-person team. Officers learned to identify anatomical landmarks that guide incision sequences during post-mortem examination.

● Understanding Wildlife Mortalities: The Veterinary Perspective

This session introduced the fundamental framework for classifying wildlife deaths — natural versus unnatural, acute versus chronic. Dr. Nigam covered ante-mortem and post-mortem changes in detail, helping officers understand how ambient temperature, sun exposure, and time since death alter the carcass. The session was particularly valued by officers who routinely encounter carcasses in advanced decomposition.

● Infectious Causes of Mortality in Elephants

Co-delivered by Dr. Karikalan Mathesh and Dr. A.B. Shrivastav, this session provided a systematic review of infectious diseases known to cause elephant mortality. EEHV received particular attention, with the critical role of whole blood PCR emphasized. FMD, Rabies, Tuberculosis, Anthrax, and Haemorrhagic Septicaemia were also covered, with participants equipped to recognize field signs and gross pathological changes indicative of each disease.

● Non-Infectious & Disaster Mortalities

Dr. Shrivastav presented a systematic framework for distinguishing electrocution deaths from other causes — characteristic lesions (entry/exit burns, supra-orbital lesions, internal haemorrhages) and the investigative report's role in triggering action against illegal power lines. Lightning strikes, train collisions, drowning, and infighting deaths were also covered with emphasis on distinguishing pathological features.

● Non-Infectious Diseases in Elephants

The final Day 1 session addressed toxicological causes — a critically under-investigated category in the Indian context. Dr. Karikalan presented detailed information on mycotoxicosis, with particular attention to kodo millet toxicity (a documented cause of elephant deaths in central India), and reviewed environmental bio-toxins. Officers received a framework for considering toxicological causes when initial investigation reveals no clear evidence of trauma, infectious disease, or natural illness.

■ ■ Session Highlights · Day 2 · 06 June 2026

● Biosafety & Biosecurity at Elephant Mortality Sites

Day 2 opened with a critical session on the biosafety risks associated with elephant carcasses. Dr. Karikalan outlined the spectrum of zoonotic risks — from anthrax and tuberculosis to other pathogens transmissible through carcass contact — and the standard PPE requirements for all personnel. The session covered safe carcass disposal, site decontamination protocols, and biosecurity measures required when infectious disease is suspected.

● Equipment Familiarization & Site Preparation

A hands-on session focused on the practical challenges of elephant necropsy under field conditions. Participants were introduced to heavy-duty equipment required — chain saws, axes, winches, block-and-tackle systems — and the sequence for their use. The critical importance of establishing a secure investigation perimeter and documenting the site before any physical intervention was emphasized.

● Comprehensive Sampling Protocols

This session addressed one of the most critical gaps in current practice: collection, preservation, and transport of biological samples from elephant carcasses. The full spectrum of sample types was covered — fresh tissue for histopathology, blood for virology, swabs for bacteriology, rumen contents for toxicology. Participants received detailed instruction on container types, preservation media, cold chain requirements, and documentation standards required for ICAR-IVRI's diagnostic laboratory.

● Recording Information During Necropsy

Dr. Karikalan and Dr. Shrivastav covered systematic documentation requirements for a complete necropsy record — including carcass condition scoring, gross pathological findings by organ system, preliminary cause-of-death assessment, and sample inventory. Guidance was provided on the specific photographs required at each stage to create a complete visual record admissible as evidence in legal proceedings.

● Vetro-Legal & Forensic Aspects of Mortality Investigation

Dr. Chandra Prakash Sharma delivered this pivotal session on legal and forensic dimensions of mortality investigation: chain-of-custody protocols, the legal framework for ivory/tusk custody under the Wildlife Protection Act, and forensic report standards that withstand court scrutiny. Participants gained clarity on their legal obligations at a mortality site and procedures for coordinating with police and revenue authorities.

● Practical: Crime Scene Investigation & Panel Discussion

Dr. Sharma led participants through a simulated crime scene investigation covering evidence marking, photography, chain-of-custody documentation, and evidence packaging. Officers practiced ivory/tusk extraction procedures and the documentation sequence that creates a legally defensible investigation record. The subsequent Panel Discussion addressed specific operational challenges raised by field officers — from accessing remote mortality sites to multi-agency coordination protocols.

■ ■ Key Outcomes & Recommendations

The two-day workshop generated concrete outcomes and forward-looking recommendations reflecting the consensus of national experts and field officers from both Van Mandals.

■ ■ Key Outcomes

- Thirty-plus field officers from Dharamjaigarh and Raigarh Van Mandals and officers from elephant-affected districts trained in systematic elephant mortality investigation.
- A standardized **Field Necropsy Checklist** and **Sample Collection Protocol** developed and distributed to all participants, adapted for Chhattisgarh's field conditions.
- Consensus reached on a **Minimum Evidence Standard** for elephant mortality investigation reports — covering carcass documentation, sample inventory, photographic record, and preliminary cause-of-death assessment.
- Strengthened institutional relationships between the Chhattisgarh Forest Department, WII, ICAR-IVRI, and NDVSU for ongoing technical support and diagnostic laboratory access.
- A framework for a **dedicated Elephant Mortality Investigation Cell** within the Bilaspur Circle proposed, with defined roles, equipment requirements, and referral pathways.

■ ■ Recommendations

Standard Operating Procedure

The Chhattisgarh Forest Department should develop and circulate a formal departmental SOP for elephant mortality investigation, drawing on the protocols shared during the workshop and adapted to the specific terrain, resources, and legal context of the state.

Dedicated Mortality Investigation Team

Each Van Mandal should designate a core team of trained officers — including at least one Range Officer with necropsy training — as the first-response unit for elephant mortality sites, with a pre-staged equipment kit maintained at division level.

Laboratory Linkage with ICAR-IVRI

A formal sample referral arrangement with ICAR-IVRI, Bareilly, should be established to ensure that biological samples from elephant mortalities receive priority processing and systematic diagnostic reporting back to the field department.

Annual Refresher Training

Given the complexity of elephant mortality investigation and the regular turnover of field staff, an annual refresher training programme is strongly recommended — potentially co-facilitated by trained Chhattisgarh Forest Department officers.

Electrocution Prevention as Priority

With electrocution accounting for 51% of elephant deaths in the study period, the workshop strongly endorses prioritizing identification and removal of illegal and low-hung power lines as the single most impactful intervention available.

■ ■ Valedictory Session

06 June 2026 · 16:45 – 17:30 hrs · Raigarh, Chhattisgarh

The valedictory session brought the two-day workshop to a formal and reflective close. Presided over by the CCF, Bilaspur Circle, and attended by all senior officers and resource persons, the session provided an opportunity for participants and faculty to reflect on the learning journey of the preceding two days.

■ ■ Participant Reflections

Selected field officers from both Van Mandals shared their key learnings. Recurring themes included: the transformative clarity gained on the necropsy procedure and its legal significance; new appreciation for the forensic value of initial site documentation; and the confidence gained from understanding disease frameworks — particularly around EEHV, electrocution lesions, and toxicological causes. Several officers expressed that the practical crime scene investigation exercise was among the most valuable sessions of their careers.

■ ■ Closing Remarks by Faculty

Dr. A. B. Shrivastav expressed that the quality of engagement from Chhattisgarh's field officers was exceptional. He recommended that future editions include a live necropsy demonstration and called for the workshop to become an annual event. Dr. Parag Nigam emphasized that the most important next step was institutional — converting the workshop's learning into formal departmental protocols. He offered WII's ongoing technical support for the development of a Chhattisgarh SOP and for future training iterations.

■ ■ Closing Address

The closing address by Shri Manoj Kumar Pandey, CCF Bilaspur Circle, highlighted the significance of the workshop in Chhattisgarh's elephant conservation trajectory. He expressed gratitude to Hon'ble Forest Minister Shri Kedar Kashyap for his vision in initiating this workshop, to PCCF Shri Arun Kumar Pandey for his leadership, and to all five resource persons for their dedicated engagement over two intensive days.

What we have built in these two days is not just knowledge — it is the foundation of a system. The 48 elephant deaths we have analyzed here must be the last generation that go uninvestigated. From this workshop forward, every death in our forests will be accounted for.

— Closing Address, Valedictory Session, 06 June 2026

Participation certificates were presented to all field officers by the senior faculty and departmental officers. A group photograph of all participants and faculty was taken in closing. The workshop formally concluded at 17:30 hrs on 06 June 2026.

CHAPTER 1

Elephant Mortality Analysis in Chhattisgarh (2021–2026)

Trends, Causes, Demographics & Priority Recommendations

Chhattisgarh Forest Department | Technical Support: WII Dehradun

1. Elephant Population Status in Chhattisgarh

Chhattisgarh has witnessed a remarkable expansion of its elephant population over the past two decades. Elephants, originally transient from Odisha and Jharkhand, have established resident breeding populations across two Forest Divisions: **Dharamjaigarh (DH)** and **Raigarh (RG)**. The population has grown from just 24 individuals in 2001 to an estimated **451 in 2026**, representing a Compound Annual Growth Rate (CAGR) of approximately 13%.

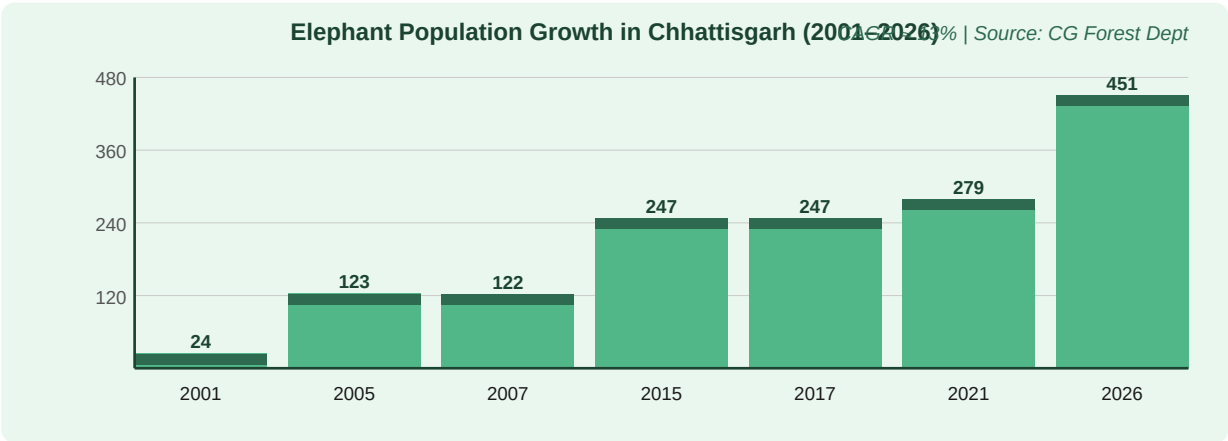


Figure 1: Elephant population growth in Chhattisgarh (2001–2026). Bilaspur Circle holds the highest concentration.

Year	Population Count	Remarks
2001	24	First recorded transient elephants from Odisha
2005	123	Rapid establishment of resident herds
2007	122	Stable — consolidation phase
2015	247	Continued range expansion into new areas
2017	247	Stable count; human-elephant conflict rising
2021	279	Post-COVID survey — increased mortality recorded
2026	~451	Estimated — highest ever; detailed census ongoing

2. Elephant Casualty Analysis (2021–2026)

TOTAL: 49 Elephant Deaths | DH Division: 27 | RG Division: 22 | 2025-26: 12 deaths (RECORD HIGH) — Immediate intervention required

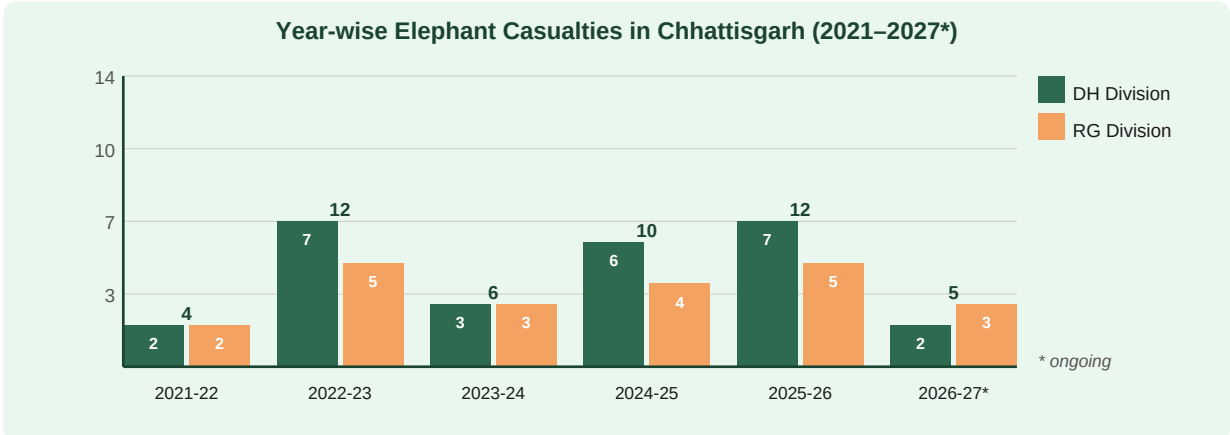


Figure 2: Year-wise elephant casualties by division (DH = Dharamjaigarh, RG = Raigarh). * 2026-27 data as of June 2026 (ongoing year).

Year	DH Division	RG Division	Total	Remarks
2021-22	2	2	4	Baseline year — low mortality
2022-23	7	5	12	Surge in deaths — monsoon related
2023-24	3	3	6	Relatively low year
2024-25	6	4	10	Rising trend resumes
2025-26	7	5	12	Record high — electrocution predominant
2026-27*	2	3	5*	Ongoing year (as of June 2026)
TOTAL	27	22	49	Cumulative 2021-2026

Human Casualties in Human-Elephant Conflict

Alongside elephant deaths, the ongoing human-elephant conflict has resulted in **45 human deaths** during 2021–2026 (DH: 36 deaths | RG: 9 deaths). This underscores the urgency of conflict mitigation alongside mortality investigation capacity.

3. Cause of Death Analysis

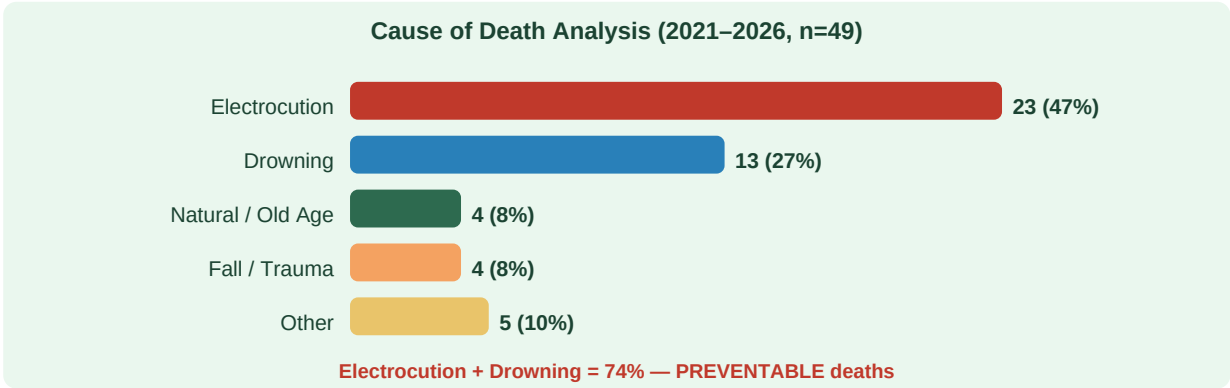


Figure 3: Cause of death distribution (n=49). Electrocutation and drowning account for 74% of all deaths — both are largely preventable with targeted interventions.

Cause of Death	Count	Percentage	Key Mechanism
Electrocutation	23	47%	Stray power lines, illegal electric fencing around crops
Drowning	13	27%	Calves unable to exit water bodies; steep banks, dams
Natural / Old Age	4	8%	Multi-molar loss, senility — natural attrition
Fall / Trauma	4	8%	Steep terrain, mine shafts, road/rail conflict
Other / Undetermined	5	10%	Toxicology pending, disease, inter-elephant conflict

4. Demographic Analysis of Casualties

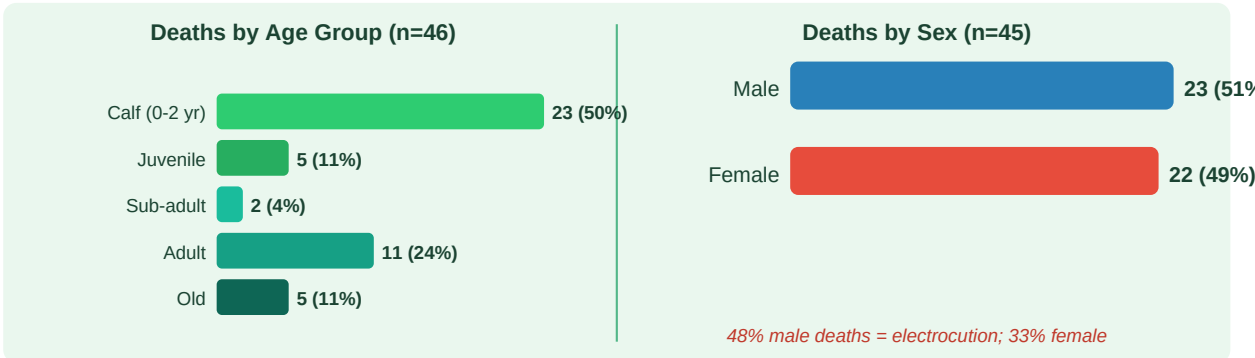


Figure 4: Age group (left) and sex (right) distribution of casualties (n=46 and n=45 respectively). Calves are disproportionately affected — 50% of all deaths.

Age Category	Count	%	Key Vulnerability
Calf (0–2 yr)	23	50%	Drowning (calves cannot exit steep banks), electrocution
Juvenile (2–5 yr)	5	11%	Separation from herd, foraging near crop fields
Sub-adult (5–15 yr)	2	4%	Dispersing males — increased exposure to power lines
Adult (15–40 yr)	11	24%	Electrocution in bulls; old cow mortality
Old (>40 yr)	5	11%	Natural senility, molar exhaustion, starvation

CRITICAL: Calves (0-2 yr) account for 50% of all deaths and 92% of all drowning deaths. Drowning prevention infrastructure

5. Seasonal Variation in Mortality

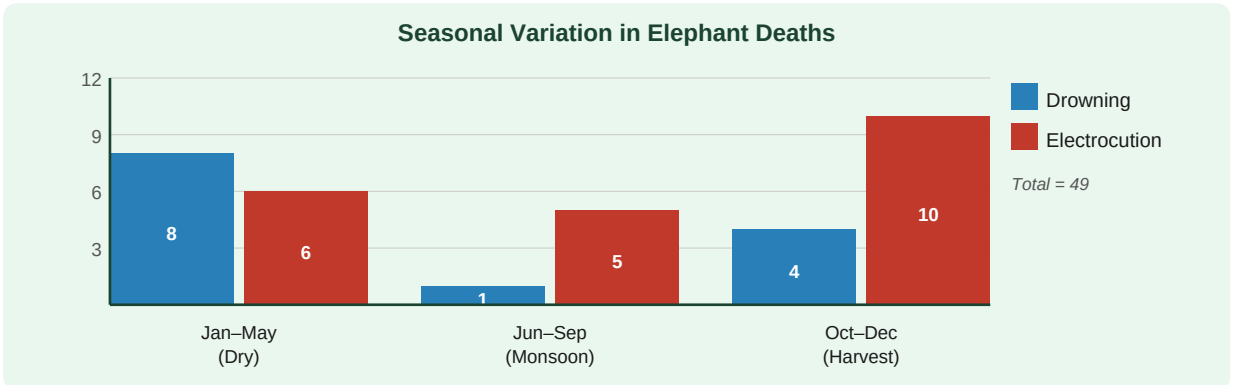


Figure 5: Seasonal distribution of deaths by cause. October is the single deadliest month (8 deaths). Oct–Dec harvest season accounts for 42% of annual deaths.

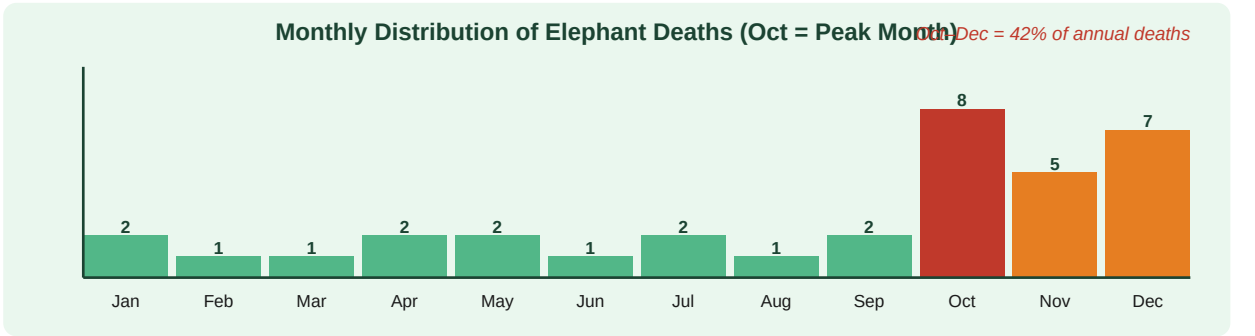


Figure 6: Monthly distribution of all elephant deaths (2021–2026 combined). October–December forms a clear mortality peak driven by illegal electric fencing during harvest.

Season	Months	Drowning	Electrocution	Key Driver
Dry Season	Jan–May	8	6	Low water levels — calves trapped in drying pools / tanks
Monsoon	Jun–Sep	1	5	Flooded rivers, reduced visibility of power lines
Harvest Season	Oct–Dec	4	10	PEAK: Illegal electric fencing to protect paddy/maize crops
TOTAL	—	13	21*	*excludes 2 unconfirmed electrocution deaths

6. Range-wise Distribution of Deaths

Analysis of death locations reveals distinct spatial hotspots that require targeted management interventions. Two ranges account for 59% of all deaths and warrant immediate priority action.

Forest Range	Division	Elec.	Drown.	Other	Total	Status
Ghargoda	RG	8	5	2	15	CRITICAL
Chhal	DH	7	5	2	14	CRITICAL
Dharamjaigarh	DH	3	2	1	6	HIGH
Kharsia	RG	2	1	1	4	MODERATE
Tamnar	RG	2	1	1	4	MODERATE
Other Ranges	—	1	—	5	6	Monitor
TOTAL	—	23	13	13	49	—

Human Conflict Hotspots

Range	Human Deaths	Priority Status
Chhal	10	HIGHEST PRIORITY — targeted patrol + power audit
Borojh	9	HIGH — community early warning system needed
Lailunga	8	HIGH — corridor mapping and fencing required
Total (45 deaths)	45	Comprehensive conflict mitigation plan required

7. Suspected Drowning — Post-Mortem Analysis

KEY FINDING: Drowning = 13 deaths (27% of total). 92% of drowning victims were calves. ARSENIC DETECTED in 2 drownings.

All 13 suspected drowning deaths underwent systematic post-mortem examination. Consistent PM findings confirmed asphyxia/drowning as the primary cause. Critically, diatom testing was positive in all tested cases. Two specimens from Tamnar Range (Dec 2025) returned positive for arsenic (File No. F.2-19/DI/NRC/2025-26/CWL), indicating possible water body contamination requiring environmental follow-up.

PM Finding	Observation	Significance
Frothy discharge	Present in ALL 13 cases	Hallmark of antemortem drowning — not seen in post-mortem drowning
Waterlogged lungs	Present in ALL 13 cases	Lung weight 3–5× normal; confirms inhalation of water
Water in trachea	Present in ALL 13 cases	Rules out post-mortem artefact — water aspirated while alive
GI water content	Large volume — ALL cases	Swallowed water while struggling — corroborates active drowning
Diatom test	POSITIVE — all tested cases	Diatoms in bone marrow = systemic dissemination = confirmed antemortem drowning
Arsenic	DETECTED in 2 specimens	Tamnar Range, Dec 2025 — environmental contamination investigation initiated

EEHV	NEGATIVE all cases	Viral haemorrhagic disease ruled out
Nitrate-Nitrite	NEGATIVE all cases	Fertiliser/industrial contamination ruled out
Heavy metals (Pb/Hg/Cd)	NEGATIVE all cases	Common industrial contaminants not implicated
Organochlorine/phosphate	NEGATIVE all cases	Pesticide poisoning excluded
HCN (Cyanide)	NEGATIVE all cases	Plant/industrial cyanide toxicity excluded

8. Case Study — Gurda Calf Drowning (01 June 2026)

Parameter	Details
Date	01 June 2026
Location	Gurda River sandbank, Kharsia Range, Raigarh Van Mandal
GPS Coordinates	22.0439°N, 83.1313°E
Victim	Male calf, estimated age 12–18 months
Discovery	Calf found separated from herd on exposed sandbank after overnight flooding
Circumstantial evidence	River flash flood event; steep eroded banks at GPS location; no other herd members present at recovery
PM Findings	Waterlogged lungs, frothy tracheal discharge, water in stomach (large volume), no external trauma, no injuries consistent with predation or electrocution
Diatom Test	POSITIVE — Naviculaceae spp. detected in lung and bone marrow
Cause of Death	Confirmed drowning (asphyxia due to water inhalation)
Follow-up	Water sample collected from Gurda River for arsenic/chemical analysis; site flagged for earthen ramp installation

9. Key Recommendations & Priority Action Plan

Based on the 5-year mortality analysis, the following eight priority actions were identified by the expert panel as immediate interventions for the Chhattisgarh Forest Department:

#	Priority Action	Timeline	Responsible Agency
1	Environmental Water Sampling: Comprehensive toxicology sampling of Rabo and Panikshet dams and water bodies in Tamnar Range for arsenic and heavy metals	Immediate	CG Forest Dept + ICAR-IVRI
2	Drowning Prevention Infrastructure: Install earthen ramps and exit pathways at all identified drowning hotspots; priority: Ghargoda and Chhal ranges	1–3 months	CG Forest Dept + PWD
3	Power-Line Audit: Complete audit and rectification of illegal electric fencing and stray HV lines across all elephant corridors; special focus on Kharsia, Ghargoda and Dharamjaigarh	1–6 months	CG Forest Dept + CSPDCL
4	Oct–Dec Enforcement Campaign: Deploy anti-poaching/anti-fencing squads during harvest season with weekly intelligence gathering in conflict-prone villages	Seasonal	Forest Dept + Police
5	Jan–May Calf Monitoring: Intensive calf surveillance during dry season when calves are most vulnerable to drowning in drying water bodies	Seasonal	WII + CG Forest Dept
6	Mandatory PM within 24 hrs: Standard Operating Procedure mandating complete post-mortem examination with sample collection within 24 hours of elephant death, with digital reporting to State WL Office	Policy — immediate	PCCF (WL), CG
7	Annual EEHV Surveillance: Annual serological and PCR surveillance for Elephant Endotheliotropic Herpesvirus across all elephant herds in DH and RG Divisions	Annual	ICAR-IVRI + WII

8	Quarterly Review Meetings: Quarterly inter-departmental review of mortality data, action plan implementation, and hotspot management with representation from Forest Dept, Police, Revenue, and Power Department	Quarterly	PCCF (WL), CG — All Depts
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Summary Statistics at a Glance

Metric	Value	Metric	Value
Study Period	2021–2026 (5 yrs)	Total Elephant Deaths	49
DH Division Deaths	27 (55%)	RG Division Deaths	22 (45%)
Electrocution Deaths	23 (47%)	Drowning Deaths	13 (27%)
Calf Deaths (0-2 yr)	23 of 46 (50%)	Male Deaths	23 of 45 (51%)
Peak Death Month	October (8 deaths)	Peak Year	2022-23 & 2025-26 (12 each)
Drowning: Calves	12/13 = 92%	Arsenic Positives	2 specimens (Tamnar, Dec 2025)
Human Deaths (HEC)	45 (DH: 36 RG: 9)	Pop. Growth (2001-26)	24 → 451 (CAGR ~13%)

Conclusion: The 5-year mortality analysis clearly demonstrates that elephant deaths in Chhattisgarh are overwhelmingly anthropogenic and preventable. Electrocution (47%) and drowning (27%) together account for three-quarters of all deaths. Calf mortality is disproportionately high at 50%. The record mortality in 2025-26 is a critical signal requiring immediate multi-agency intervention. This workshop provides the technical foundation for standardised investigation, evidence-based reporting, and preventive action.

CHAPTER 2

Biological & Anatomical Aspects of the Asian Elephant

Workshop Reference Chapter · Workshop on Essentials for Mortality Investigation of Asian Elephant · June 2026

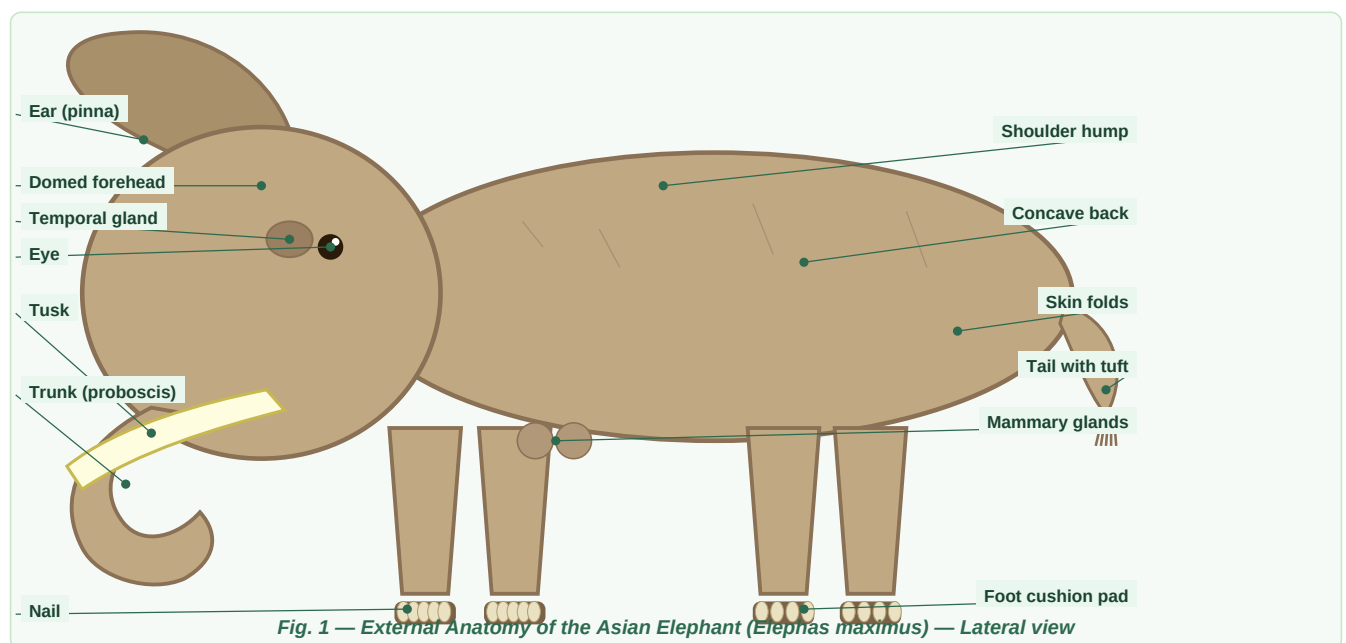
Workshop on Essentials for Mortality Investigation of Asian Elephant · Chhattisgarh Forest Department · June 2026

1. Taxonomy & Classification

The Asian elephant (*Elephas maximus*) is the largest land animal in Asia and a keystone species of tropical and subtropical forest ecosystems. It belongs to Order **Proboscidea**, the only surviving order of a once-diverse lineage of trunk-bearing mammals that included mammoths and mastodons.

Rank	Classification
Kingdom	Animalia
Phylum	Chordata
Class	Mammalia
Order	Proboscidea
Family	Elephantidae
Genus	Elephas
Species	<i>Elephas maximus</i> (Linnaeus, 1758)
Subspecies	<i>E. m. maximus</i> (Sri Lanka) · <i>E. m. indicus</i> (Indian mainland) · <i>E. m. sumatranus</i> (Sumatra) · <i>E. m. borneensis</i> (Borneo)

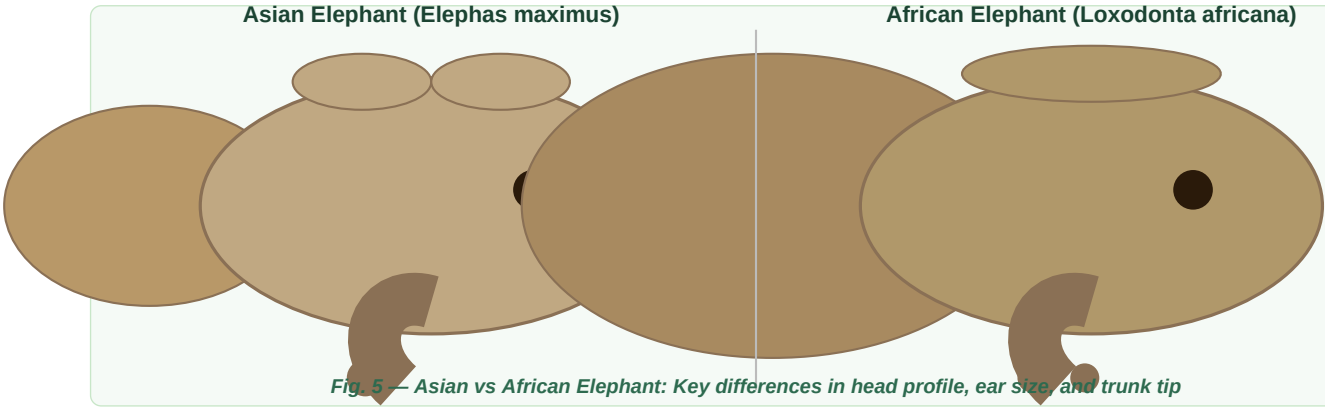
2. External Anatomy



Physical Characteristics

Parameter	Value
Shoulder height	Male: 2.5–3.5 m Female: 2.0–2.5 m
Body weight	Male: 3,500–5,000 kg Female: 2,000–3,500 kg
Body length (nose to tail)	5.5–6.5 m (male)
Skin thickness	Up to 2.5–3 cm on back/head; thinner on ears
Lifespan	60–70 years in the wild
Gestation period	~22 months (longest of any land mammal)
Inter-birth interval	3–5 years
Daily food intake	100–300 kg of vegetation
Daily water intake	100–200 litres

3. Asian vs. African Elephant — Key Differences



Feature	Asian Elephant	African Elephant
Head profile	Two-domed (divided forehead)	Single rounded dome
Ear size	Smaller (~1/3 of African)	Very large (ear = map of Africa)
Back profile	Concave / dipped	Concave / more arched
Highest body point	Shoulder	Back / mid-spine
Trunk tip fingers	1 (upper lip only)	2 (upper and lower lip)
Tusk presence	Males mainly; females often tuskless	Both sexes commonly tusked
Front foot nails	5	4–5
Hind foot nails	4	3–4
Skin texture	More wrinkled; pinkish depigmented patches common	Smoother, uniformly dark
Ear thermoregulation	Moderate (smaller pinna surface)	Very efficient (large radiator surface)

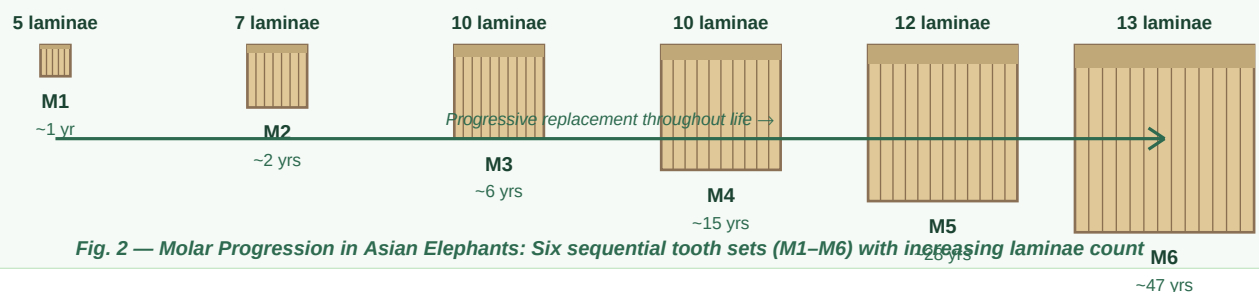
4. The Trunk (Proboscis)

The trunk is the most anatomically complex and functionally versatile organ in the animal kingdom — a **muscular hydrostat** (like the human tongue) containing approximately **100,000 individual muscle units** but

no bone whatsoever. It is a fusion of the elongated nose and upper lip.

Function	Details
Breathing	Primary airway; can breathe under water by using as a snorkel
Drinking	Sucks up 8–10 L per draw; transfers to mouth or sprays onto body
Bathing & cooling	Sprays water or mud over back and sides for evaporative cooling
Olfaction	Raised to detect scents at height; most important sensory function (up to 2,000 olfactory receptor genes)
Grasping & foraging	1 "finger" at tip (Asian) allows precise picking of leaves, fruits, and grass; can uproot trees with trunk-and-foot leverage
Vocalisation	Produces trumpeting calls (alarm, greeting, excitement); low-frequency rumbles for infrasound communication
Social bonding	Touching, intertwining, and exploring other elephants; used in greeting and consolation
Defence & combat	Striking, throwing, and pushing; males use in musth conflicts
Digging	Excavates water holes in dry riverbeds; extracts salt licks
Disease sampling	Trunk wash: key clinical sample for TB and EEHV diagnosis

5. Tusks, Teeth & Dentition



Tusks

- Tusks are **modified second upper incisor teeth** — they grow continuously throughout the elephant's life from deep sockets in the premaxillary bone.
- Composition:** solid dentine (ivory) with a small pulp cavity at the root; the outer surface has a thin enamel cap that wears off early in life.
- In Asian elephants, **only adult males** normally carry large visible tusks. Females usually have small **tushes** (rudimentary tusks) or none at all.
- Males of the **Sri Lankan subspecies** (*E. m. maximus*) are frequently tuskless — a genetically and selectively driven trait.
- Permanent tusks **protrude beyond the lips at ~30 months** and are used for digging, debarking trees, combat, and as resting props for the trunk.

Molar Dentition

Elephants are **unique among mammals** in producing six successive sets of molar teeth (M1–M6) that erupt and move forward (mesiodentally) along the jaw during life. Each molar is larger than the previous and contains more **laminae** (transverse enamel ridges that provide grinding surface). When M6 is exhausted — typically at 55–65 years — the animal can no longer chew adequately and eventually dies of malnutrition.

6. Ears & Thermoregulation

The elephant ear is a masterpiece of biological engineering — a **vascularised thermal radiator**. A dense network of superficial blood vessels runs through the ear's thin skin, and rapid flapping creates a convective airflow that dissipates heat from the blood before it returns to the core.

Aspect	Details
Vascular network	Dense capillary plexus visible through thin ear skin — blood cooled by 3–9 °C before returning to core circulation.
Ear flapping	Slow rhythmic flapping (1–3 times/min at rest; faster in heat) acts as a bio-fan, dramatically increasing convective heat loss.
Surface area	Asian elephant ear $\approx$ 0.3–0.5 m ² per ear; African ear $\approx$ 1.5–2.0 m ² — this explains why African elephants tolerate greater heat.
Communication	Ear posture signals mood: forward/spread = alert/threatening; flat against neck = submission or contentment.
Infrasound coupling	Ground vibrations from infrasound calls may be sensed through the feet AND detected via the jaw/skull; ear anatomy plays a secondary role.

7. Skin

Property	Detail
Thickness	Up to 2.5–3 cm on the back and head; much thinner (1–2 mm) on the ears
Colour	Dark grey-brown; depigmented pink-white patches common around ears, trunk, and face (especially in Asian elephants)
Texture	Heavily wrinkled and folded — wrinkles increase surface area for evaporative water loss; reduce surface temperature
Sweat glands	Absent except for small glands between the toes; transepidermal water loss through permeable skin is the main mechanism
Thermoregulation	Skin permeability increases in heat — effective when drinking water available; mud and water bathing critical
Health indicators	Cracked, dry, or peeling skin: dehydration or nutritional deficiency. Pox lesions, skin tumours, cutaneous filariasis all externally visible

8. Feet & Locomotion

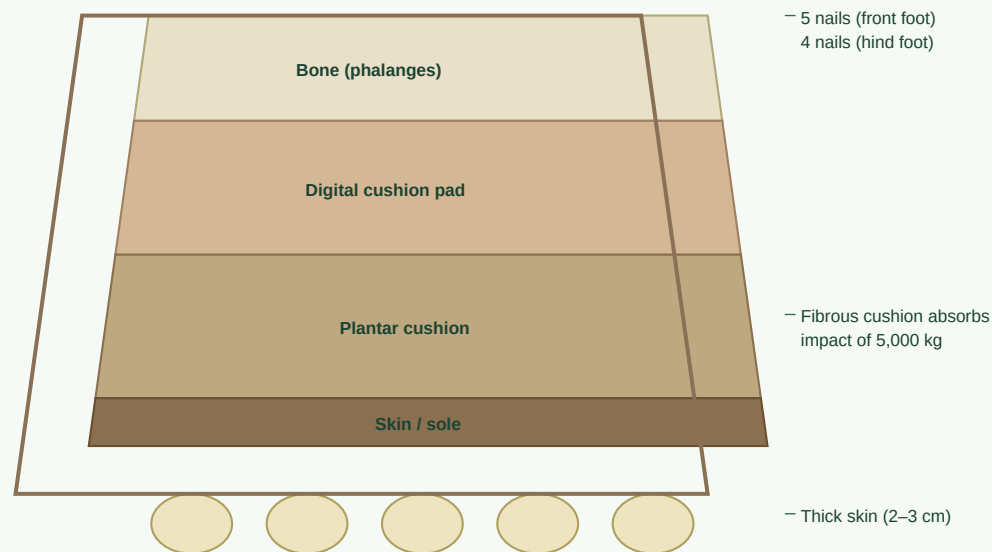


Fig. 3 — Cross-Section of the Asian Elephant Foot (showing cushion pad layers and nails)

- **Semi-digitigrade posture** — elephants walk on the tips of their digits, with a large fibrous cushion pad (like a built-in heel pad) absorbing impact.
- The cushion pad acts as a **hydraulic shock absorber**, distributing the 3,500–5,000 kg body weight across the foot to reduce pressure per cm².
- Despite enormous weight, elephants walk **almost silently** — the cushion pad muffles footsteps, essential for stealthy forest movement.
- **Maximum speed:** ~25 km/h (elephants do not truly gallop; all four feet never leave the ground simultaneously).
- **Asian elephant nails:** 5 on the front foot, 4 on the hind foot. Overgrown, cracked nails are a common captive management problem.
- Footpad health is critical — infections or injuries in foot pads can be life-threatening for captive elephants.

9. Skeletal System

Structure	Feature
Cervical vertebrae	7 (same as all mammals)
Rib pairs	20 pairs (most land mammals have 12–13)
Lumbar vertebrae	3–4 (fewer than most mammals; aids weight-bearing)
Bone density	Pachyosteosclerosis — bones are extremely dense and compact, providing structural support for massive weight
Femur & humerus	Vertically positioned (columnar) — unlike most quadrupeds; entire leg acts as a column, not a pendulum
Skull	Large but surprisingly light — numerous air-filled diploe (cancellous bone chambers) reduce skull weight
Foot bones	All five digits present; digit bones (phalanges) short; sesamoid bones within the cushion pad
Tusk attachment	Deep root in premaxillary bone; periodontal ligament allows slight flexibility during use

10. Digestive System

100–300 kg food/day · 100–200 L water/day · ~40% digestive efficiency · 24–48 hr GI transit

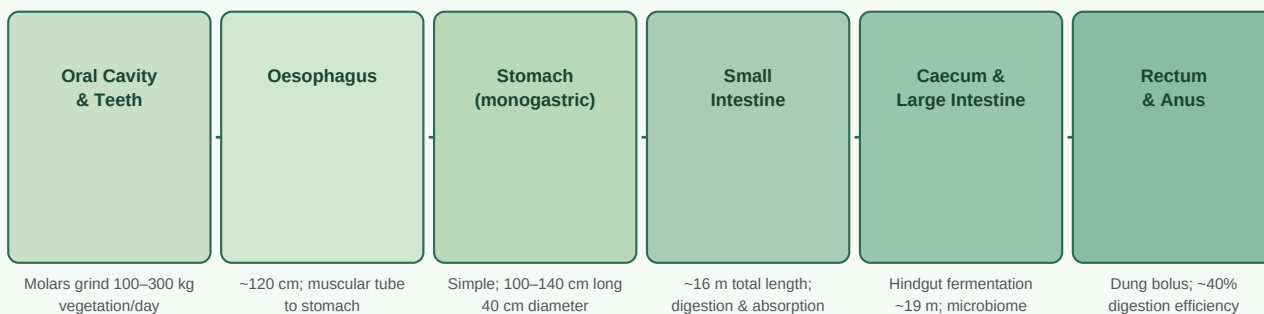


Fig. 4 — Digestive System of the Asian Elephant (Hindgut Fermenter — Monogastric)

- **Monogastric (simple-stomached)** — unlike ruminants, elephants have a single-chambered stomach (~100–140 cm long, ~40 cm diameter).
- **Hindgut fermentation** — microbial breakdown of cellulose occurs primarily in the enlarged caecum and colon (not in the stomach/small intestine).
- **Relatively low digestive efficiency (~40%)** — compensated by the enormous volume consumed (100–300 kg/day).
- **Total intestinal length** can reach ~35 metres (stomach + small + large intestine + caecum + rectum).
- **GI transit time:** 24–48 hours — undigested seeds dispersed over large distances (elephants are keystone seed dispersers).
- **Clinical relevance:** sudden feed changes → gut microbiome disruption → colic, diarrhoea, or increased toxin production.

11. Reproductive System

Feature	Detail
Mammary glands	Located between the forelimbs (pectoral position) — unique among large mammals; two glands with one teat each
Oestrous cycle	~16 weeks; ovulation associated with two LH surges
Gestation	~22 months (longest of any land mammal)
Age at first calving	Females: ~14 years; males not usually successful until >20 years
Inter-birth interval	3–5 years (limited by extended lactation)
Sexual maturity (male)	~15 years; socially mature at 25–30 years
Musth	Annual period of heightened testosterone (10–100× normal) in adult males; temporal gland secretion, urine dribbling, aggressive behaviour; lasts days to months
Temporal gland	Paired exocrine glands between eye and ear; active during musth and in females during oestrus; examined at necropsy
Testes	Internally located (abdominal); not externally visible
Penis	S-shaped when erect; males urinate backward
Placentation	Zonary haemochorial; relatively large placenta

12. Brain, Senses & Communication

Brain

- The Asian elephant brain weighs **~4–5 kg** — the largest brain of any land mammal in absolute terms.
- Highly convoluted cortex; well-developed **hippocampus** (memory) and **temporal lobe** (social cognition).
- Demonstrated capabilities include: **mirror self-recognition**, **tool use**, empathy, altruism, and long-term memory (>20 years).
- Encephalisation Quotient (EQ) is lower than primates but higher than most other large mammals.

Sense	Mechanism & Notes	Capability
Olfaction (smell)	Most dominant sense. >40% of genome dedicated to olfactory receptors — more than any other animal studied. Trunk raised to sample scent plumes at height.	Exceptional (>2,000 OR genes)
Infrasound hearing	Produce and detect calls at 14–35 Hz (below human hearing range). Seismic vibrations also sensed through feet and jaw.	Communication up to 10 km
Tactile sense	Trunk tip is exquisitely sensitive (Meissner's corpuscles). Sensitive skin on feet detects ground vibrations.	Excellent
Vision	Relatively poor compared to other senses. Dichromatic colour vision (blue-yellow; red appears as green). Better in dim light than bright sun.	Limited; poor in bright light
Taste	Well-developed; use to select preferred food plants and detect contaminants or toxic plants.	Good

Key Takeaways

Chapter Summary — Biological & Anatomical Aspects of the Asian Elephant

1. Asian elephants (*Elephas maximus*) are identified by their double-domed head, smaller ears, concave back, and single trunk-tip finger — key distinctions from the African elephant.
2. The trunk is a boneless, 100,000-muscle organ critical for survival; trunk wash is the priority clinical sample for EEHV and TB diagnosis.
3. Six successive molar sets (M1–M6) allow age estimation; exhaustion of M6 at 55–65 years leads to eventual death from inability to chew.
4. The foot cushion pad silently absorbs enormous weight; nails and pads are critical health indicators in captive animals.
5. Hindgut fermentation in the caecum/colon with low (~40%) efficiency is compensated by enormous daily food intake (100–300 kg).
6. 22-month gestation, 3–5 year inter-birth interval, and slow sexual maturity make the population extremely sensitive to adult mortality.
7. Olfaction (>2,000 olfactory receptor genes) and infrasound communication (detectable up to 10 km) are the dominant senses.

CHAPTER 3

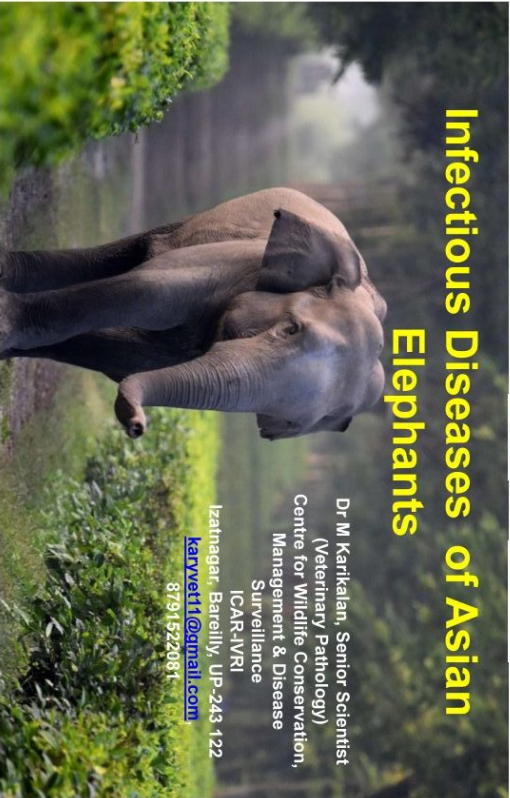
Infectious Diseases of Asian Elephants

Resource Person: Dr. M. Karikalan, MVSc, PhD

Assistant Professor, Dept. of Wildlife Science, VBSC & AH, Namakkal

Workshop on Elephant Mortality Investigation and Management

Chhattisgarh Forest Department · Bilaspur · June 2026



Infectious Diseases of Asian Elephants

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The need to conserve !!

- Current population status of Asian elephants shows a declining trend (Williams *et al.*, 2020)
- Asian elephant categorised as "Endangered" by IUCN
- Assigned under Schedule 1 of wildlife protection act 1972, conferring highest protection
- Listed as Appendix 1 species by CITES
- Population of Asian elephants in India as per the 10th census (2010) was 42,446 elephants in the wild (Purcell & Co., 2025)

Habitat loss and fragmentation

- Illegal trade
- Poaching
- Human elephant conflict

(Sakumar *et al.*, 2016; Meena *et al.*, 2017)

National heritage animal of India

Classification

Kingdom	Animalia
Phylum	Chordata
Class	Mammalia
Order	Proboscidea
Family	Elephantidae
Subfamily	Elephantinae
Genus	Loxodonta
Species	Loxodonta cyclotis (African forest elephant)
Genus	Loxodonta
Species	Loxodonta africana (African bush elephant)
Genus	Elephas
Species	Elephas maximus (Asian elephant)
Subspecies	Elephas maximus sumatranus
	Elephas maximus indicus
	Elephas maximus maximus
	(Roca <i>et al.</i> , 2015)

Source presentation: Infectious Diseases of Asian Elephants (Dr. M. Karikalan, 2026)

Overview

- **Species:** Asian Elephant (*Elephas maximus*) — Endangered (IUCN)
- **Wild population:** ~22,446 individuals across 13 range countries
- **India:** Largest population; Schedule I (Wildlife Protection Act 1972); CITES Appendix I
- **Scope of lecture:** Infectious, parasitic and management-related diseases — diagnosis, treatment and prevention in wild and captive populations
- **Lecture series:** VBSC SH_1_3 — compiled and presented for field veterinarians and forest officers

1. Introduction

Wildlife disease investigation has emerged as a critical pillar of conservation medicine. The Asian elephant, already classified as Endangered on the IUCN Red List, faces an escalating burden of infectious and non-infectious diseases, many of which are poorly understood in wild populations. Mortality from preventable or treatable diseases represents a significant threat that can be mitigated through timely diagnosis, skilled post-mortem examination, and preparedness of field veterinarians.

1.1 Epidemiological Terminology

A clear understanding of disease epidemiology is essential for fieldwork. Key terms used throughout this chapter are defined below:

Term	Definition
Emerging Disease	A disease appearing in a population for the first time, or rapidly increasing in incidence/geographic range
Re-emerging Disease	A previously known disease that has reappeared after a period of decline, often in new geographic areas or host species
Transboundary Disease	Infectious disease that spreads across national or ecosystem boundaries, requiring multi-jurisdictional response (e.g., FMD, HS)
Zoonosis	Disease transmissible between vertebrate animals and humans (e.g., TB, Rabies, Anthrax, Leptospirosis)
Spillover	Transmission of a pathogen from a reservoir host to a new (incidental) host species
Pathogenicity	Ability of a pathogen to cause disease in the host
Virulence	Severity of disease caused by a pathogen; degree of harm inflicted on the host

1.2 Taxonomy of the Asian Elephant

Kingdom	Animalia
Phylum	Chordata
Class	Mammalia
Order	Proboscidea
Family	Elephantidae
Genus	Elephas
Species	Elephas maximus
Subspecies	E. m. indicus (Indian), E. m. maximus (Sri Lankan), E. m. sumatranus (Sumatran), E. m. borneensis (Borneo)

1.3 Spectrum of Disease

Elephant diseases are broadly categorised into infectious and non-infectious conditions. This distinction is fundamental for triage at a casualty site — while infectious conditions require diagnostic sampling and biosafety precautions, non-infectious causes (trauma, electrocution, nutritional deficiency) require different response protocols.

Infectious Diseases	Non-Infectious Conditions
Viral: EEHV, Rabies, FMD, Poxvirus, EMCV	Electrocution (illegal/accidental)
Bacterial: Tuberculosis, Haemorrhagic Septicaemia, Anthrax, Clostridial infections, Leptospirosis	Drowning / Water body falls
Parasitic: Helminths (Fasciola, Strongyles, Amphistomes), Protozoa (Balantidium, Haemoparasites)	Trauma / Fall from height
Fungal: Aspergillosis (captive)	Obesity and nutritional deficiency
	Inter-elephant conflict / musth-related injury
	Captive management failures (chaining, overwork)

2. Elephant Endotheliotropic Herpesvirus (EEHV)

EEHV is the single most important viral disease in elephants. It causes acute haemorrhagic disease characterised by a peracute clinical course — an apparently healthy calf can die within 24–72 hours of first signs. Understanding EEHV biology, diagnosis, and treatment is therefore a priority competency for any field or zoo veterinarian working with elephants.

2.1 Classification and Subtypes

EEHV belongs to the family *Herpesviridae*, subfamily *Betaherpesvirinae*. In Asian elephants, the clinically important subtypes are:

EEHV Subtypes Relevant to Asian Elephants

- **EEHV-1A / 1B** — Most virulent; responsible for majority of fatal cases in Asia and North America
- **EEHV-4** — Second most common in Asian elephants; documented fatal cases
- **EEHV-5** — Occasionally detected; less frequently fatal
- **EEHV-2, 3, 6, 7** — Primarily documented in African elephants; lower risk in Asian species
- **Carrier state:** Adult elephants are latent carriers with virus persisting in nerve ganglia; disease triggers in calves when maternal antibody wanes (between 1–18 years of age)

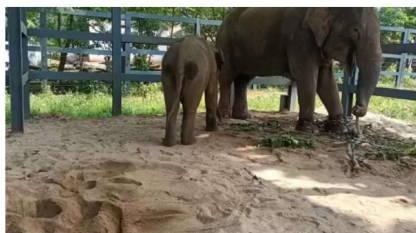
Figure 2.1 — Herpesvirus family classification and EEHV phylogenetic tree

Clinical signs observed

- Lethargy/ restlessness
- Rise in body temperature
- Reduced appetite
- Salivation
- Swelling of temporal gland and eye lids
- Cyanosis of tongue
- Abdominal colic
- Recumbency

Death with in 6 to 24 hours

Pic courtesy: Nandankanan Zoo Authorities

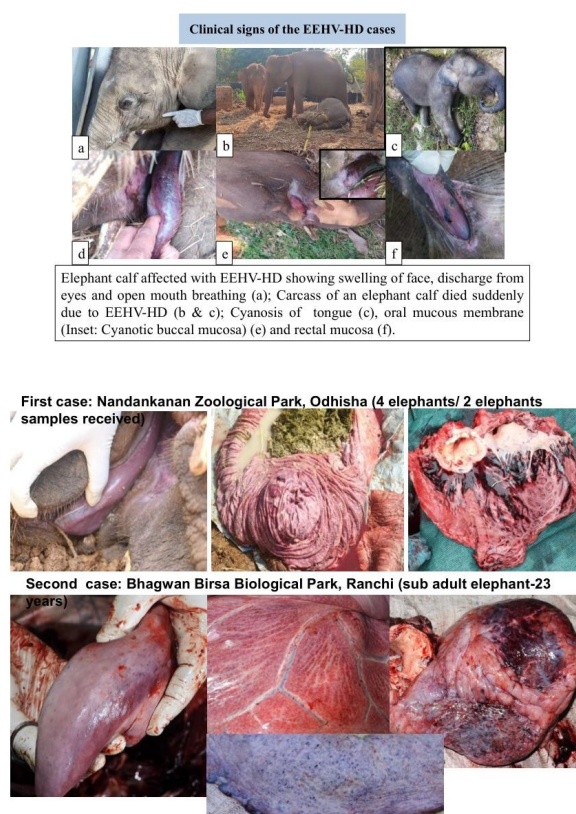


2.2 Historical Timeline and India Cases

EEHV haemorrhagic disease (EEHV-HD) was first recognised in 1995 at the Smithsonian National Zoo, Washington DC, in an Asian elephant calf. Since then, multiple fatal cases have been documented globally and in India:

Year	Event / Location
1995	First EEHV-HD case recognised — Smithsonian National Zoo, USA
2007	First confirmed EEHV case in India — Kerala (retrospective diagnosis)
2008–2015	Cases documented in Tamil Nadu (Anamalai Tiger Reserve, Mudumalai)
2021	Chhattisgarh: EEHV-1A positive detected (qPCR from calf death)
2023	Chhattisgarh: Second positive case — active surveillance programme initiated
2024	mRNA-based EEHV vaccine enters trial phase — Houston Zoo, USA

Figure 2.2 — EEHV phylogenetic relationships and fatal haemorrhagic disease history



2.3 Pathomechanism and Vulnerable Age Groups

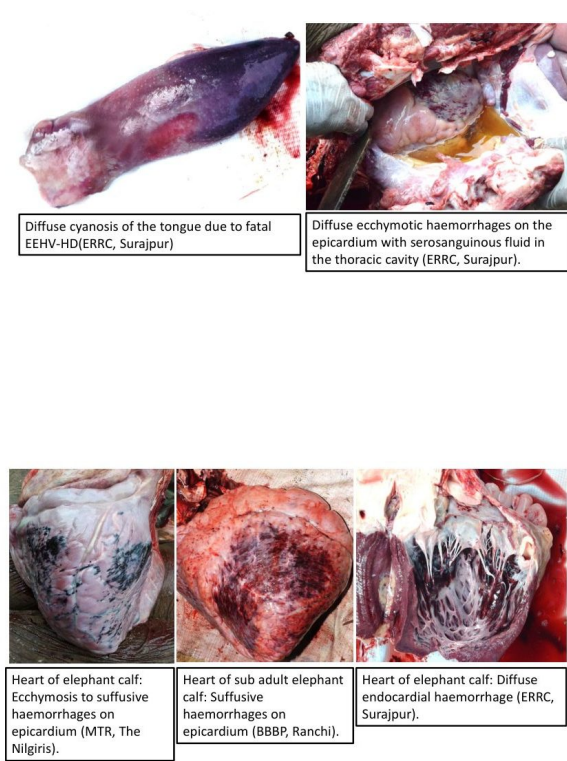
The host preference of EEHV is age-dependent. Calves between 1 and 18 years are the primary victims of fatal disease. Adult elephants harbour the virus as latent carriers but rarely develop clinical disease. The pathomechanism involves endothelial cell tropism — the virus infects cells lining blood vessels, causing massive vascular damage:

Pathomechanism Cascade

- 1. Viral entry:** EEHV infects endothelial cells of capillaries and venules

2. **Vascular damage:** Endothelial lysis → capillary leakage → haemorrhage into tissues
3. **Cytokine storm:** Systemic inflammatory response → hypotension → shock
4. **Organ failure:** Myocardial, hepatic and renal involvement → death within hours to days
5. **Gross lesions:** Epicardial haemorrhages, tongue cyanosis, oedema of face/tongue, petechiation

Figure 2.3 — EEHV host preference by age group and pathomechanism cascade



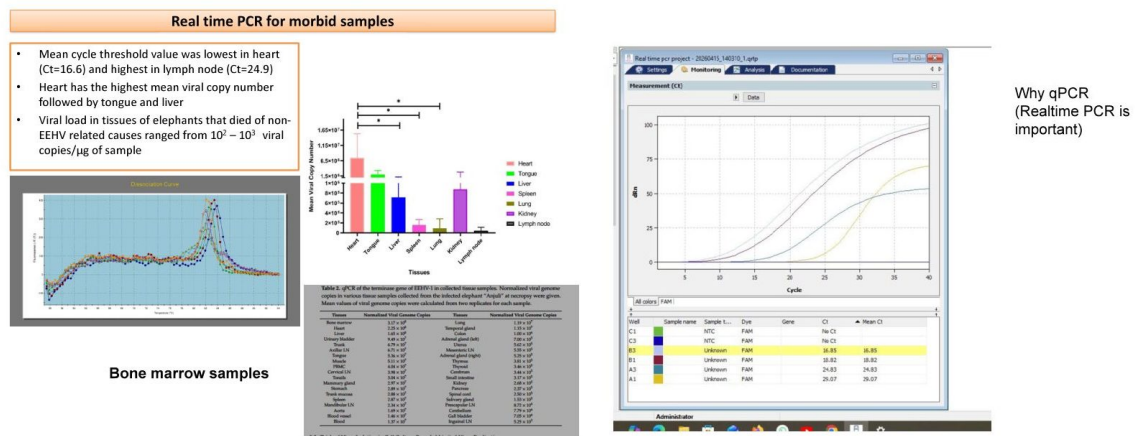
2.4 Clinical Signs

EEHV-HD presents as a peracute haemorrhagic syndrome. The time from first clinical signs to death may be as short as 12–36 hours. The following signs have been documented in confirmed cases:

Clinical Signs of EEHV-HD

- **Generalised oedema** of the head, face, and neck — especially dramatic periorbital and submandibular swelling (hallmark sign)
- **Tongue cyanosis:** Blue-purple discolouration of the tongue mucosa (visible in live animal)
- **Open-mouth breathing** and respiratory distress due to oropharyngeal oedema
- **Weakness, recumbency,** reluctance to walk — rapid deterioration to lateral recumbency
- **Pyrexia** (fever $>38.5^{\circ}\text{C}$) often the first sign detected during routine morning observation
- **Tachycardia** and hypotension in late stages
- **Bloody discharge** from natural orifices may occur terminally
- Death within 24–72 hours of first signs in untreated or late-detected cases

Figure 2.4 — Clinical photographs: EEHV-HD in a calf (Nandankanan Zoological Park)



Left: Facial/periorbital oedema in affected calf. Right: Recumbency and open-mouth breathing.

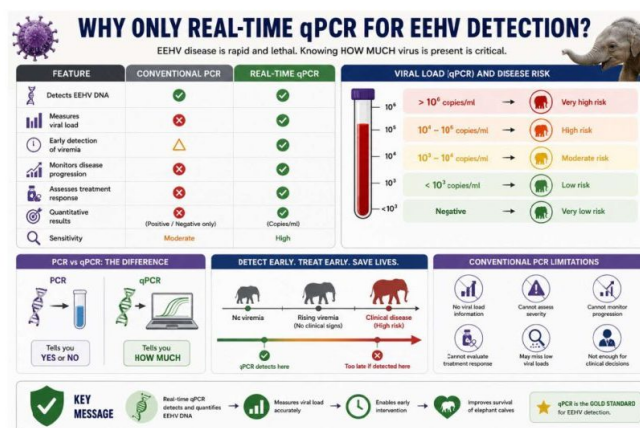
2.5 Post-mortem Findings

Necropsy of EEHV-HD cases reveals characteristic gross and histopathological changes. A thorough and documented post-mortem examination is essential for laboratory confirmation and to rule out other causes of sudden death in young elephants:

Gross Post-Mortem Lesions

- **Tongue:** Diffuse cyanosis/necrosis; villous thickening; dark blue-black discolouration on cut section
- **Heart:** Epicardial and endocardial petechiae and ecchymoses; myocardial pallor and mottling
- **Liver:** Congestion; occasional petechiae; hepatocellular swelling
- **Lung:** Congestion and oedema; interstitial haemorrhage; frothy exudate in airways
- **Lymph nodes:** Markedly enlarged, haemorrhagic; mesenteric LN especially affected
- **Kidney:** Cortical pallor, perivascular haemorrhage
- **Spleen:** Usually unremarkable or mildly enlarged
- **Brain:** Usually negative (important for rabies differential exclusion)

Figure 2.5 — PM photographs: tongue cyanosis (ERRC Surajpur), epicardial haemorrhages (MTR Nilgiris), and myocardial lesions (BBBP Ranchi, Nandankanan)



Treatment of EEHV-HD

- Start immediately without waiting for laboratory confirmation
- Human antiviral drugs : used to treat animals with clinical illness (Schmitt *et al.*, 2000; Cracknell, 2008; Wiedner *et al.*, 2012)
- Per-rectal administration of famciclovir @5.5-8.0 mg/kg BID (Schmitt *et al.*, 2001; Tociłowski *et al.*, 2015; Howard *et al.*, 2015; Long *et al.*, 2015; Fuery *et al.*, 2016; Fuery *et al.*, 2016)
- Initial IV administration followed by oral therapy of ganciclovir @5mg/kg twice daily (Stanton *et al.*, 2013; Padilla *et al.*, 2015)
- Acyclovir @12mg/kg BID for 15 days along and penicillin G (Boonprasert *et al.*, 2015)
- Oral acyclovir therapy (Khammeeri *et al.*, 2020)
- Famciclovir @ 8-15 mg/ kg body weight QID on 1 st day and then BID for 3 weeks orally or rectally (Famciclovir is the first drug of choice)
- **Fluid therapy**
 - IV and rectal administration of fluid
 - Per rectal fluid administration: 10-15 litres depending on the clinical status of the animal @ 20ml per kg body weight
 - Plasma therapy
- Diuretics: Management of peripheral edema :Furosemide 0.6 mg/ kg bw, IV or IM QID
- Antibiotics, anti-inflammatory drugs and immuno-modulators(Levamisole-75 S/C (single dose @ 1ml/50 kg), Vitamin-C)



2.6 Histopathology

Histopathological examination confirms the endotheliotropic nature of EEHV. Representative tissues should be fixed in 10% neutral buffered formalin immediately at necropsy. Characteristic

findings on H&E; staining include:

Histopathological Findings (H&E;)

- **Tongue mucosa:** Intranuclear inclusion bodies in endothelial cells; mucosal ulceration; sub-epithelial haemorrhage; intravascular fibrin thrombi
- **Myocardium:** Perivascular lymphocytic infiltration; myocyte necrosis; endothelial swelling with inclusion bodies (H&E; ×200)
- **Pulmonary sinusoids:** Endothelial inclusions; alveolar capillary congestion; occasional fibrin thrombi
- **Renal vessels:** Glomerular and interstitial haemorrhage; endothelial inclusions
- **Diagnostic significance:** Intranuclear inclusions are pathognomonic for herpesvirus; their presence in multiple tissues confirms EEHV-HD

Figure 2.6 — Histopathology: H&E; sections (×200) from EEHV-HD cases

Monitoring disease, prevention and control

- Vaccines are still not available (recombinant vectored vaccines, subunit vaccines??!!)
- New or rescued animal should be properly quarantined before mixing with an existing herd.
- Animal should be quarantined for a period of 6 months at a separate facility with periodical screening of blood sample and trunk wash by PCR besides regular monitoring of vital signs
- Regular close monitoring of animal, visual inspection of different parts of the body, vital signs and behavioral pattern
- CBC, serum biochemical analysis to assess any abnormal haematological changes like leucocytosis, monocytosis, lymphopenia and thrombocytopenia
- Blood sample and trunk wash should be screened at weekly interval against virus through PCR to suspect/confirmed positive elephant
- Screening protocol should be in place for all zoo/private/camp having captive elephant to identify carrier/ shedder in herd
- In pregnant females only periodical monitoring of viral load through q-PCR from the samples like conjunctival swab/ trunk wash/ vaginal wash may be carried out at three-month intervals
- Proper biosecurity measures

ZOO NEWS BLOG

Houston Zoo Elephant Receives First Ever mRNA EEHV Vaccine

June 18, 2024



Tess, a 40-year-old Asian elephant at Houston Zoo, has been given the first-ever dose of an mRNA vaccine created by virologists at Baylor College of Medicine (BCM) to prevent the deadly elephant endotheliotropic herpesvirus (EEHV) IA—a devastating viral disease in Asian elephants worldwide. Over the next few weeks, animal care experts and veterinary staff at Houston Zoo will closely monitor Tess to learn how effective this vaccine is for elephants.

mRNA vaccines
Recombinant vaccines

Clockwise from top-left: Tongue mucosa, myocardium, pulmonary sinusoids, renal vessels — showing intranuclear inclusion bodies and vascular damage (H&E; ×200)

2.7 Laboratory Diagnosis: qPCR

Quantitative PCR (qPCR) is the gold standard for EEHV diagnosis, both in live animals (antemortem) and at necropsy (postmortem). Understanding the CT value interpretation is essential for field veterinarians:

Sample Type	Use	Interpretation
Whole blood (EDTA)	Antemortem surveillance; monitoring in sick calves	CT <30: Active viraemia (treat immediately); CT 30–37: Low-level positive / carrier; CT >37: Negative or below detection
Bone marrow aspirate	Best postmortem sample; highest viral load in active EEHV-HD	Highest sensitivity — preferred for PM diagnosis
Trunk wash	Detection of viral shedding in carriers/adults	Used for herd surveillance; lower sensitivity than blood PCR
Heart / Tongue / Lymph node	PM tissue samples for qPCR and histopathology	Heart and tongue show highest viral copy numbers in EEHV-HD
Liver / Lung / Kidney	Supplementary PM tissues	Variable viral loads; important for severity assessment

Figure 2.7 — Tissue viral load data (qPCR) and qPCR vs conventional PCR comparison

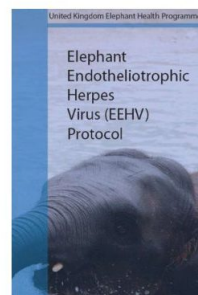
Samples: Blood, serum, trunk wash fluids, oral and rectal swab, tissues, Bone marrow

Standard Operating Procedure for Elephant Endotheliotropic Herpes Virus - Hemorrhagic Disease



CENTRAL ZOO AUTHORITY
Statutory body of the Ministry of Environment, Forest and Climate Change,
Government of India

2021



- IVRI, Izatnagar, Bareilly
- Veterinary College, Guwahati
- Centre for Wildlife Health, OUAT, Bhubaneswar,
- Centre for Wildlife Forensic & Health, Jabalpur
- EEHV Diagnostic laboratory, Kerala Forest Department, Wayanad
- State Institute For Animal Diseases, Palode, Kerala

Rabies

- ❑ Rabies is a viral zoonotic neuro invasive disease which causes inflammation in the brain and is usually fatal
- ❑ Transmission mainly through wild canines, Rodents such as mice , rats, squirrels and chipmunks and **small carnivores (Mongoose)**
- ❑ All warm blooded animals are susceptible .
- ❑ The virus can not live outside its host for more than a couple of seconds but has been found in animals as long as 48 hrs after death.
- ❑ In India this disease was mainly reported in captive wild animals of various zoos like **Asian elephants (both in captive and free range)**.
- ❑ Transmitted mainly through the bite from infected animals.
- ❑ Virus may be present in saliva before the onset of clinical signs
- ❑ Variable IPd of 1 week to 1 year
- ❑ Dumb or the furious form
- ❑ Death occurs within 10 days of the 1st symptom

2.8 Treatment Protocol

EEHV-HD is a veterinary emergency. Treatment must begin immediately on clinical suspicion — do not wait for PCR results. Early intervention dramatically improves survival odds. The treatment protocol used at leading elephant centres in India:

EEHV Treatment Protocol (Field & Zoo)

- **Famciclovir** 5.5–8 mg/kg BID orally — antiviral prodrug (converted to penciclovir); inhibits viral DNA polymerase; first-line agent; must be pre-positioned at elephant camps
- **Acyclovir** IV — used in severe cases when oral famciclovir cannot be administered; acts at same target as famciclovir; give via slow IV infusion
- **Fluid therapy:** Isotonic crystalloids (Ringer's Lactate, Normal Saline) via IV; volume resuscitation for vascular leakage; monitor urine output
- **Anti-inflammatory:** Dexamethasone (cautious use) or Meloxicam for inflammatory cascade
- **Gastric protectants:** Omeprazole / Pantoprazole — prevent stress-related gastric lesions
- **Supportive care:** Shade, quiet environment, soft browse/fruit, mahout presence to reduce stress
- **Monitoring:** Temperature every 4 hours; blood qPCR every 24–48 hours; decreasing CT value = improving; CBC and chemistry every 48 hours if possible

CRITICAL: EEHV Emergency Response

Famciclovir and Acyclovir must be pre-positioned at all elephant camps and zoos housing calves. Delay in starting antiviral treatment is the most common cause of treatment failure.

Do NOT use corticosteroids as first-line agents — they may accelerate viral replication.

Quarantine the affected calf; test all herd members for viraemia; notify State and Central authorities.

Figure 2.8 — EEHV treatment protocol and monitoring/prevention framework



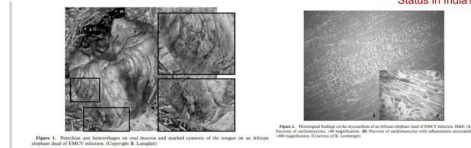
Samples sent directly to DFMD, Bhubaneswar or Muketeshwar campus

- FATAL ENCEPHALOMYOCARDITIS VIRUS INFECTION IN AN AFRICAN SAVANNA ELEPHANT (
- LOXODONTA AFRICANA*
-) IN A FRENCH ZOO

Abstract. A fatal case of encephalomyocarditis virus (EMCV) involving an African elephant (*Loxodonta africana*) occurred in November 2011 at the Réserve Africaine de Yagoua, France. An adult female was found dead without any preliminary symptoms. Gross pathological changes consisted of petechiae and hemorrhages on mucosae and internal organs, abundant transudate in the abdominal and pericardial cavities, and myocarditis. Histopathological examination revealed a severe myocarditis with a mixed infiltrate of mononuclear cells and concurrent lymphoplasmacytic and eosinophilic infiltrate. An EMCV was isolated from several organs and considered the causative agent of the myocarditis. The same strain of virus was also isolated in rodents captured on new premises and considered to be the reservoir of the virus. To the authors' knowledge, this is the first EMCV in a giraffe (*Giraffa camelopardalis*) and the first in an African elephant.

Key words: African elephant, death, encephalomyocarditis, *Loxodonta africana*, virus, zoo.

Status in India?



Workshop on Elephant Mortality Investigation and Management · 2026 · Page 13

3. Rabies in Elephants

Rabies, caused by *Lyssavirus* (negative-sense ssRNA, family *Rhabdoviridae*), is a fatal zoonotic neurological disease. While uncommon in elephants, confirmed and suspected cases have been documented in captive individuals, typically following contact with infected dogs, foxes, or bats. The invariably fatal outcome and zoonotic risk make it a priority differential diagnosis in any elephant with acute neurological signs.

3.1 Transmission and Clinical Features

- **Transmission:** Bite of a rabid dog (most common in captive elephants in India); rarely from fox or bat
- **Incubation:** Variable (weeks to months); virus travels along peripheral nerves to CNS
- **Clinical signs:** Sudden behavioural change; aggression alternating with depression; hypersalivation; dysphagia; photophobia; incoordination; seizures; coma; death
- **Differential diagnosis:** EEHV (young calves), HS, OP poisoning, musth-related aggression
- **Prognosis:** 100% fatal once clinical signs appear; no treatment available

3.2 Diagnosis

Rabies Diagnostic Protocol

- **Sample:** Brain tissue (cerebellum, hippocampus, brainstem) — ONLY sample for rabies diagnosis
- **FAT (Fluorescent Antibody Test):** Gold standard; requires fresh brain; fluorescent microscopy
- **Histopathology:** Negri bodies — eosinophilic intracytoplasmic inclusions in Purkinje cells and hippocampal neurons (H&E; ×400)
- **Seller's stain:** Rapid field stain for Negri bodies (methylene blue + basic fuchsin); Negri bodies appear magenta against blue background (×1000)
- **RT-PCR:** Most sensitive; gel electrophoresis shows diagnostic band; can use preserved tissue
- **Biosafety:** Handle brain samples with gloves, mask, and eye protection; post-exposure prophylaxis for exposed personnel

Figure 3.1 — Rabies: transmission cycle, Negri body histology and diagnostic methods

Delhi Zoo's African Elephant Died Of Viral Infection, Reveals Post-Mortem

Officials said the Encephalomyocarditis (EMCV) virus, which causes inflammation of heart muscles, was identified as the cause of death. The elephant was found dead in its enclosure on September 17.

Press Trust of India [Delhi News](#) Nov 04, 2025 07:00 am IST

Read Time: 2 mins

Share



Top row: Rabies transmission cycle; H&E; ×400 — Negri bodies in neurons; Seller's stain ×1000. Bottom row: FAT (Fluorescent Antibody Test) smear; RT-PCR gel showing diagnostic band.

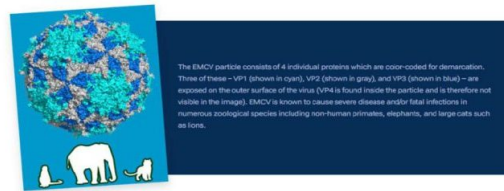
4. Foot and Mouth Disease (FMD)

FMD is caused by *Aphthovirus* (Picornaviridae), with seven known serotypes (O, A, C, SAT1, SAT2, SAT3, Asia1). While elephants are not primary hosts, spillover from infected livestock is documented — particularly in forest fringes where elephant corridors overlap with grazing areas. FMD rarely causes death in elephants, but severe cases with secondary septicaemia can be fatal.

FMD in Elephants — Key Points

- **Clinical signs:** Vesicular lesions on feet and trunk tip; lameness; reluctance to walk; drooling if oral lesions present; fever in acute phase
- **Lesion appearance:** Fluid-filled blisters (vesicles) → rupture → raw erosions → secondary bacterial infection
- **Diagnosis:** Clinical appearance of vesicles; virus isolation; RT-PCR from vesicular fluid; ELISA for serology; probe-based antigen detection kit (IVRI Mukteswar)
- **Control:** FMD vaccination of all surrounding livestock; strict movement restrictions; lime disinfection of premises; isolation of affected animals
- **Significance in elephants:** Captive Indian elephants reported with FMD; secondary septicaemia is the main cause of mortality risk

Figure 4.1 — FMD vesicular lesions and isolation protocol



Encephalomyocarditis virus (EMCV; family Picornaviridae) is a rodent-borne virus that is very well-known to the zoological community worldwide, as it has repeatedly caused severe outbreaks resulting in the deaths of zoo mammals – most notably elephants – for decades. We have been investigating recent EMCV outbreaks in zoos in the U.S. that have resulted in mortality of numerous African and Asian mammals including elephants, hippos, and multiple primate species, including those that are endangered. Understandably, these outbreaks and the loss of unique and rare megafauna are a major concern to zoos.

The main objectives of this project are to:

- Identify the sources and drivers of recent outbreaks of EMCV
- Characterize the evolutionary history of EMCV in wildlife
- Delineate how surface mutations in EMCV affect autogenous vaccine efficacy

As EMCV outbreaks will undoubtedly continue to occur in the future, new avenues of vaccine research and a better understanding of the natural history of the virus are critically needed in order to adequately protect animals from fatal disease. Ultimately, these studies will allow us to identify factors that facilitate EMCV emergence in zoos and how EMCV genetic diversity affects antigenicity and future vaccine development.

POX

- Cowpox virus infections have been reported as a cause of a severe, sometimes fatal disease in Asian elephants in European zoos.
- African elephants can be affected as well, but usually the lesions in this species are restricted to the skin and they tend to heal quickly.
- The causative virus is an orthopox virus, affecting rodents (endemic hosts), elephants etc.
- It causes local or generalized vesicular lesions of the skin and mucous membranes and can develop into a systemic disease, affecting the lungs and GI-tract. Intra-uterine infection in an Asian elephant has been reported.
- Pox infection in elephants is a zoonotic disease and has been reported in human caretakers after contact with affected elephants.



Cowpox lesions in an old Asian elephant.



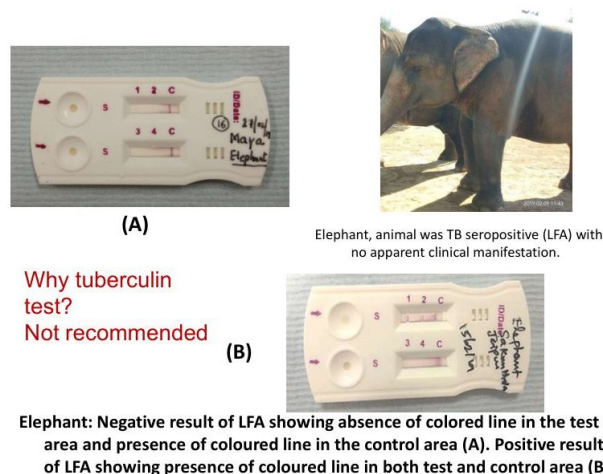
5. Poxvirus and Encephalomyocarditis Virus (EMCV)

5.1 Poxvirus (Cowpox/Elephantpox)

Elephants can be infected with cowpox virus (a member of *Orthopoxvirus*), typically through contact with infected rodents or dairy cattle. The condition is sometimes called "elephantpox" but is genetically cowpox virus. Cases are documented in zoos across Europe and in India.

- **Lesions:** Papular/vesicular eruptions on skin — trunk, ears, periocular region, feet; heal over 2–4 weeks in uncomplicated cases; gross lung lesions in severe systemic pox
- **Diagnosis:** Clinical appearance; electron microscopy of skin scrapings; PCR; histopathology (H&E; eosinophilic intracytoplasmic inclusions — Guarnieri bodies)
- **Treatment:** Symptomatic; wound care; antibiotics for secondary infection; tecovirimat (antiviral) in severe cases
- **Zoonosis:** Cowpox is a zoonotic virus; mahouts and keepers handling lesions must use PPE

Figure 5.1 — Poxvirus: vesicular skin lesions on Asian elephant



5.2 Encephalomyocarditis Virus (EMCV)

EMCV (Cardiovirus, Picornaviridae) causes acute myocarditis and sudden cardiac death, predominantly in zoo elephants. The virus is transmitted via rodent contamination of food and water. A fatal case was documented at Delhi Zoo (India). The virus replicates in myocardial cells, causing haemorrhagic myocarditis.

- **Source:** Rats and mice (reservoir hosts); contaminated feed and water troughs
- **Clinical signs:** Sudden death or acute cardiac failure; weakness; collapse; occasionally preceded by brief respiratory distress
- **PM findings:** Haemorrhagic myocarditis; pale myocardium; pericardial effusion
- **Diagnosis:** Virus isolation; serology (ELISA); histopathology; rodent control as prevention

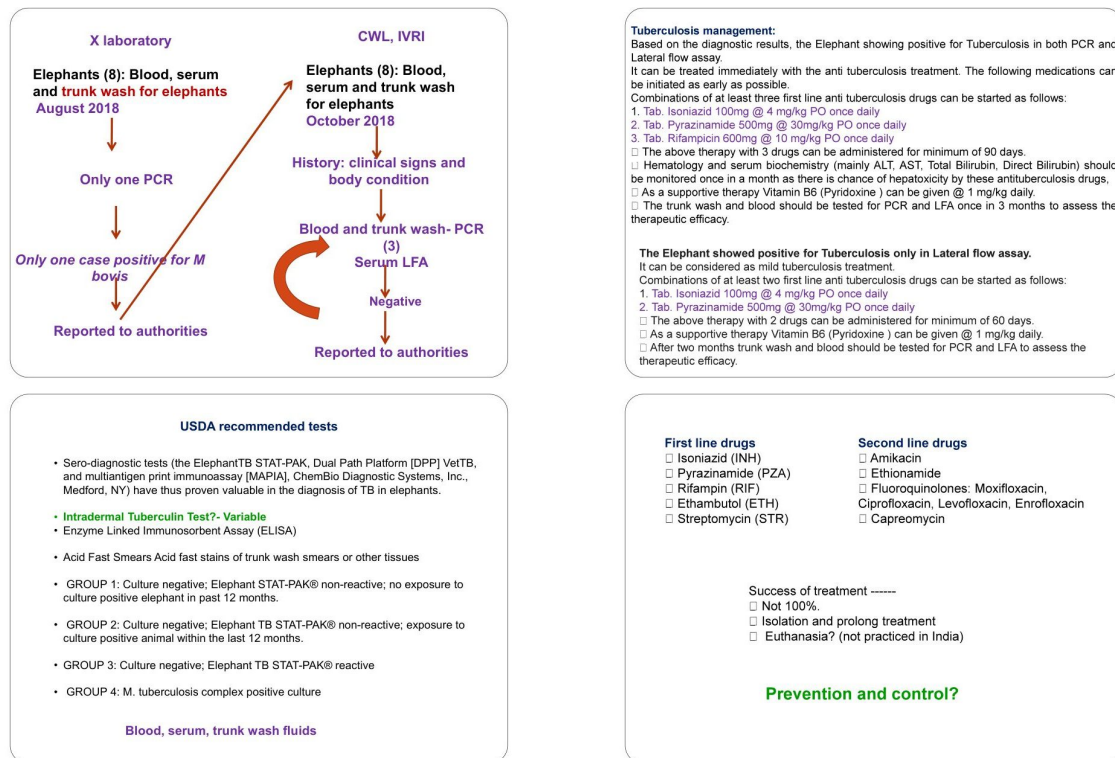
6. Tuberculosis (TB) in Elephants

Tuberculosis caused by *Mycobacterium tuberculosis* (human strain) is the primary mycobacterial disease in captive elephants globally. Importantly, TB in elephants is a reverse zoonosis — humans with active TB infect elephants, not vice versa in most documented cases. However, infected elephants can shed bacilli and theoretically transmit to in-contact humans (mahouts, vets). The disease is chronic, slowly progressive, and often clinically silent for years.

6.1 Diagnosis

TB Diagnostic Protocol (CWL IVRI)
Trunk wash (TW):
• Standard antemortem test — 3 trunk washes (saline lavage) on 3 consecutive mornings; pooled sample • Culture on Lowenstein-Jensen medium (takes 6–8 weeks); PCR (faster — 48–72 hours) • Sensitivity 60–70% for culture; PCR more sensitive but needs validation
Serology (MAPIA / ElephantTB STAT-PAK):
• Multi-Antigen Print ImmunoAssay; lateral flow assay using recombinant antigens • Detects antibodies — useful for screening but cannot distinguish active from latent disease • Serial testing increases sensitivity; single negative test does not rule out TB
PM diagnosis:
• Gross miliary TB lesions in lung, lymph nodes (acid-fast staining, Ziehl-Neelsen) • PCR on lymph node tissue; IS6110 insertion sequence target (<i>M. tuberculosis</i> specific)

Figure 6.1 — TB diagnostics: acid-fast staining, PCR gel and LFA kit



Left: Acid-fast bacilli (blue stain) in sputum/trunk wash; IS6110 PCR gel; LFA kit (TB STAT-PAK). Right: USDA recommended diagnostic flowchart; free-range elephant lung with miliary TB lesions.

6.2 Treatment

TB Treatment Protocol (Anti-Tubercular Therapy)

• 3-drug regimen (minimum 12 months; ideally 18–24 months for elephants):

— Isoniazid (INH): 100 mg/day or 4–5 mg/kg

— Pyrazinamide (PZA): 500 mg/day or 25–30 mg/kg (first 2 months)

— Rifampicin (RIF): 600 mg/day or 10–15 mg/kg

• Drugs given in banana/jaggery boluses; observed administration (OAT) essential

• Monitor liver enzymes (AST, ALT) monthly — hepatotoxicity risk

• Pyridoxine (Vitamin B6) supplementation to prevent INH-related neuropathy

• Restrict movement and contact with other elephants during treatment

TB: Regulatory and Biosafety Requirements

TB in elephants is a notifiable disease. Report to Chief Wildlife Warden and nearest IVRI/veterinary institute.

Mahouts caring for TB-positive elephants must undergo chest X-ray and sputum test; anti-TB prophylaxis if indicated.

Three negative trunk wash cultures at monthly intervals required before declaring an elephant TB-free.

7. Anthrax

Anthrax is caused by *Bacillus anthracis*, a Gram-positive, aerobic spore-forming bacterium. It causes peracute to acute disease in elephants, with death occurring within hours of the onset of signs. Anthrax outbreaks in wildlife are typically associated with specific anthrax-prone "hot zones" where spores persist in the soil for decades.

Anthrax — Key Field Points

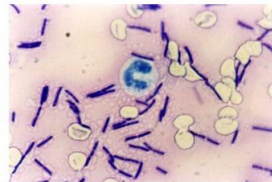
- **Peracute/acute course:** An apparently healthy elephant found dead; or brief signs of unsteadiness, collapse, death within 1–6 hours
- **Pathognomonic gross sign:** Non-clotting, tarry (dark, watery) blood from all natural orifices (mouth, trunk, anus, vulva)
- **DO NOT open the carcass** on site — sporulation occurs on air contact; opening the carcass contaminate the entire site for 30–50 years
- **Staining:** Polychrome methylene blue (McFadyean reaction) — bamboo-rod shaped bacilli with blue capsule in blood smear; pathognomonic
- **Spore persistence:** Anthrax spores (endospores) survive in soil for 30–100 years; heavy rainfall or soil disturbance can expose buried carcasses and release spores
- **Epidemics:** Outbreaks tend to occur in hot, dry seasons followed by rainfall; documented in Joyepur Forest Range, Dibrugarh, Assam (elephants and ungulates)

Figure 7.1 — Anthrax: carcass findings, polychrome methylene blue staining, and field investigation (Joyepur Forest Range, Assam)

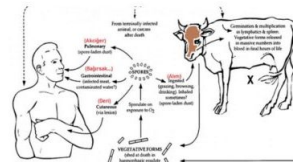
A photograph showing a dead elephant lying on its side on a dirt path. A person is standing next to the elephant, and another person is partially visible on the right. The elephant's body is on the ground, and its legs are visible. The background shows some greenery and a dirt path.



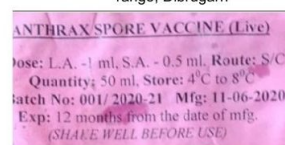
Three plasmid-coded exotoxins:
edema factor,
lethal toxin
,protective antigen



Polychrome methylene blue staining/Giemsa



Anthrax outbreak at Joyepur forest range, Dibrugarh

[illegible]

DO NOT perform a full necropsy on a suspected anthrax carcass in the field.

All persons at scene must wear full PPE: gloves, N95 mask, goggles, boot covers.

Collect only a peripheral blood smear and blood in sealed vials for lab confirmation.

Notify Chief Wildlife Warden, IVRI, and District Collector (Section 38P, WPA 1972).

Vaccinate all livestock in a 5-km radius with Sterne strain anthrax vaccine immediately.

8. Haemorrhagic Septicaemia (HS)

Haemorrhagic Septicaemia is caused by *Pasteurella multocida* type B:2 (and occasionally E:2). HS is primarily a disease of bovids (cattle, buffalo), but elephants are susceptible — particularly captive/semi-captive animals under stress. The disease is peracute and characterised by sudden high fever, respiratory distress, and death within 24–48 hours. Outbreaks in elephants have been investigated in Kalahandi district, Odisha and Kartapat WLS.

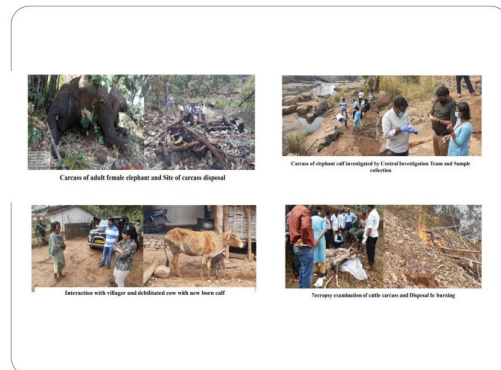
HS — Diagnosis and Management

- **Predisposing factors:** Physical stress (translocation, overwork, musth), nutritional deficiency, heavy rainfall season, contact with infected cattle
- **Clinical signs:** High fever (40–42°C); oedematous swelling of head/neck/brisket; respiratory distress; profuse salivation; tympanitis; death in 24–48 hours
- **PM findings:** Severe haemorrhagic pneumonia; peracute congestion of all organs; haemorrhagic lymph nodes; subcutaneous gelatinous oedema
- **Histopathology:** Bipolar organism (oval shape with bipolar staining — "safety pin" appearance) on Gram stain or Leishman stain
- **Best diagnostic sample:** Bone marrow aspirate from femur or rib (taken within hours of death before lysis; highest yield)
- **Treatment:** Penicillin G 20,000 IU/kg IM; Tetracycline; Sulphonamides; fluid therapy; anti-inflammatory (Meloxicam)
- **Prevention:** HS vaccination of all livestock in a 2-km radius; separate elephant water sources from cattle wallows

Figure 8.1 — HS: bipolar organisms (Leishman stain), haemorrhagic lung lesions, and Kalahandi field investigation

WHO recommendation for Anthrax carcass burning not burial

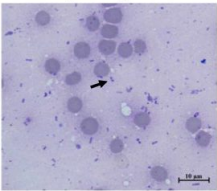
Why?



Left: Bipolar staining of *P. multocida*; gross PM lung lesions; haemorrhagic lymphadenopathy. Right: Field investigation at Kartapat WLS — trackers, necropsy team, and carcass disposal operations.

Figure 8.2 — HS control: livestock vaccination and bone marrow sampling protocol

Bipolar organisms



Control measures

1. Vaccination against HS for domestic animals
2. Carcass disposal
3. Available Samples
4. Awareness

**Heart blood, LN, Lung
Bone marrow is the ideal sample**

Molecular characterisation and serotyping of *Pasteurella multocida* isolates from Asiatic elephants (*Elephas maximus*)

2016

Chandranak, Basavegowdanadoddi Marinaik | Shivashankar, Beechagondahalli Papanna | Giridhar, Papanna | Nagaraju, Doddegowdanakoppai Narjagowda

We report the cultural and molecular characterisation of *Pasteurella multocida* isolates recovered from Asiatic elephants at Bandipur tiger reserve forest in Karnataka state, India. The forest had witnessed consecutive droughts from 2010 to 2012 and sudden heavy monsoon rains in 2013. Twenty-three elephants died during this period of heavy rains between May and July 2013. Postmortem conducted on a fresh elephant carcass revealed hemorrhagic tracheitis, haemorrhages on heart and lesions of acute septicemia in all other vital organs. In seven elephant carcasses where fresh samples were not available, bone marrow samples of femur bones were aseptically collected. Heart blood and spleen samples collected at postmortem and two out of seven bone marrow samples yielded the growth of *P. multocida*. The isolates were typed as *P. multocida* type B. The phylogenetic analysis of the isolates was carried out by 16S ribosomal RNA (rRNA) gene-based PCR and sequencing of 1.3 Kbp nucleotides on the 16S rRNA gene of the isolates. The isolates showed highest sequence identity with *P. multocida* isolates of bovine and caprine origin. This is the first report of molecular study on *P. multocida* from Asiatic elephants.

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Bone marrow and Lymph nodes

9. Clostridial Infections

Clostridial diseases are caused by Gram-positive, anaerobic, spore-forming bacilli of the genus *Clostridium*. Several species are pathogenic in elephants, producing potent toxins that cause rapid multiorgan failure:

Species	Disease	Key Features
<i>C. difficile</i>	Pseudomembranous enterocolitis	Watery/haemorrhagic diarrhoea; pseudomembrane on colonic mucosa; associated with antibiotic use
<i>C. perfringens</i> (Type A/D)	Enterotoxaemia	Sudden death; bloat; haemorrhagic enteritis; toxin detection by ELISA from intestinal contents
<i>C. tetani</i>	Tetanus	Muscular rigidity; lockjaw; opisthotonus; triggered by wound infection; rare in elephants
<i>C. botulinum</i>	Botulism	Flaccid paralysis; toxin from contaminated feed; peracute death possible
<i>C. chauvoei</i>	Blackleg (rare)	Gas gangrene of muscle; crepitation on palpation; acute febrile death

Field Note: Clostridial Sample Collection

When handling intestines of suspected Clostridial cases: do NOT open intestinal loops in the field. Intestinal contents must be sealed and sent to laboratory immediately under cold chain.

Clostridial toxins dissipate rapidly after death — intestinal samples must be collected within 1–2 hours of death for toxin ELISA to be meaningful.

Figure 9.1 — Clostridial: *C. difficile* enterocolitis (gross mucosa), tetanus, and published case literature

Clostridial infections

Enterotoxaemia,

Tetanus,

Clostridium difficile Enterocolitis

***Clostridium perfringens* type A beta2 toxin in elephant (*Elephas maximus indicus*) and pygmy hog (*Sus salvanius*) with haemorrhagic enteritis in Assam, India**

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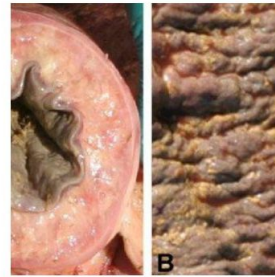
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Accepted 10 June 2019

This paper reported the investigation of haemorrhagic enteritis in female elephant (*Elephas maximus indicus*) and pygmy hog (hog) reared in the Assam State Zoo, Guwahati, Assam, India. An eight-year-old female elephant and two and half-year-old female pygmy hog developed haemorrhagic enteritis of unknown cause manifested at the age of four and four months, respectively. Bacteriological investigations revealed that the causative agent *Clostridium perfringens* was associated with the disease. Erythromycin, clindamycin and metronidazole were effective, however, ampicillin or penicillin G was more effective and probably the drug of choice for *C. perfringens* associated haemorrhagic enteritis. Isolates derived from elephant harboured four plasmids (6.1, 14.4, 36.8 and 86.2 kb), while that from pygmy hog carried two plasmids (62.8 and 91.3 kb). PCR analysis of *C. perfringens* isolates revealed presence of alpha toxin gene (*cpa*) and beta2 toxin gene (*cpb2*). None of the isolates were positive for beta1, epsilon, zeta and enterotoxin genes. The sequence analysis of partial *cpa* gene showed 98.8 to 100% homology among the isolates studied. The study confirmed the involvement of beta2 toxin-producing *C.*



Samples need to be collected and transported in anaerobic conditions

CLOSTRIDIUM INFECTIONS IN ASIAN ELEPHANTS (*Elephas maximus*)

Clostridium spp. are anaerobic, spore-forming bacteria that produce potent toxins. Spores persist in the environment and can cause severe disease in elephants.

1. Clostridium novyi

Cause: Severe tissue infection and toxemia

How it causes disease:

- Spores enter through wounds or damaged tissues.
- Germinate in anaerobic (low oxygen) environments.
- Produce toxins (alpha-toxin, beta-toxin) causing tissue destruction, edema, shock.
- May lead to sudden death.

Clinical signs:

- Swelling, painful wound
- Tissue necrosis, foul discharge
- Fever, depression, collapse
- Sudden death in acute cases

TRANSMISSION & RISK FACTORS

- Spores are widespread in soil, feces, and the environment.
- Entry through wounds, gastrointestinal tract, or antibiotics (in calves).
- Risk increases with stress, poor hygiene, injuries, antibiotic overuse, or poor diet.

2. Clostridioides difficile

Cause: Enterocolitis (inflammation of the intestines)

How it causes disease:

- Overgrowth occurs when normal gut flora is disrupted (e.g., after antibiotic use, stress, poor diet).
- Produces toxins A and B.
- Toxins damage intestinal lining, causing colitis and diarrhea.
- Leads to fluid loss, dehydration, and sometimes death.

Clinical signs:

- Watery or mucoid diarrhea
- Straining, abdominal discomfort
- Dehydration, weight loss
- Depression, weakness

DIAGNOSIS

- Clinical signs and history
- Fecal examination
- Toxin detection (ELISA)
- Culture and PCR
- Necropsy findings

3. Clostridium perfringens

Cause: Enterotoxaemia, necrotic enteritis

How it causes disease:

- Common inhabitant of the large intestine.
- Overgrowth and sudden toxin production cause intestinal damage.
- Produces toxins A and B.
- Toxins (alpha, beta, epsilon, iota, etc.) cause necrosis, gas production, and systemic toxemia.
- Can cause sudden death.

Clinical signs:

- Acute diarrhea
- Colic, abdominal pain
- Toxemia: depression, rapid breathing, weakness
- Sudden death in severe cases

TREATMENT

- Antitoxin (where available)
- Antibiotics (metronidazole, penicillin, etc.)
- Supportive care: fluids, electrolytes, NSAIDs
- Wound care and debridement

4. Clostridium tetani

Cause: Tetanus

How it causes disease:

- Spores enter through wounds.
- Germinate in anaerobic conditions.
- Produce tetanospasmin toxin.
- Toxin affects the nervous system, causing muscle rigidity and spasms.

Clinical signs:

- Muscle stiffness, rigidity
- Difficulty walking
- Trismus (locked jaw), difficulty swallowing
- Severe spasms, recumbency, may be fatal

PREVENTION

- Good hygiene and sanitation
- Proper wound care
- Judicious use of antibiotics
- Balanced diet and stress reduction
- Tetanus toxoid vaccination for at-risk elephants

Early detection and prompt treatment are critical, as clostridial diseases can progress rapidly and be fatal in elephants.

10. Leptospirosis

Leptospira spp. (spirochete bacteria) cause leptospirosis, a globally important zoonosis. The disease is transmitted through water or soil contaminated by urine of infected animals (rodents, cattle, dogs). Elephants are exposed at contaminated water bodies, river banks, and during monsoon flooding. The IVRI Leptospirosis Laboratory maintains 21 serovars for diagnosis.

- **Transmission:** Contact with contaminated water/soil through skin abrasions, mucous membranes, or ingestion; rodents are primary reservoir; cattle/dogs amplify hosts
- **Clinical signs in elephants:** Fever; icterus (jaundice); haemoglobinuria (red urine); haemorrhagic manifestations; abortion in females; in severe cases, acute renal failure and death
- **Diagnosis:** MAT (Microscopic Agglutination Test) — paired samples 2 weeks apart; 4-fold rise in titre diagnostic; Latex Agglutination Test (LAT) for field screening; darkfield microscopy of urine; PCR on blood/urine
- **Treatment:** Penicillin G (effective if early); Doxycycline 5 mg/kg; supportive fluid therapy
- **Prevention:** Avoid water body access during monsoon floods; rodent control; vaccination of working elephants (available in some countries)

Figure 10.1 — Leptospirosis: transmission cycle, MAT serovars panel, and Latex Agglutination Test

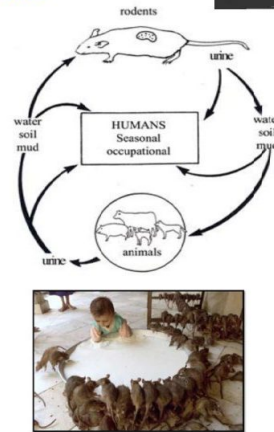
Leptospirosis

Caused by *Leptospira interrogans*

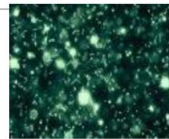
Characters:

- ▶ Thin, flexible, motile, finely coiled
- ▶ with a hook at one or both ends
- ▶ Giemsa stain: Light red
- ▶ 0.1 μm in diameter
- ▶ 6-20 μm in length
- ▶ Obligate aerobes
- ▶ Optimum temperatures of 28-30°C

- | | |
|-----------------------------------|-------------------------|
| • <i>L. autumnalis</i> , | • <i>L. hardjo</i> , |
| • <i>L. batavia</i> , | • <i>L. australis</i> , |
| • <i>L. copenhageni</i> , | • <i>L. canicola</i> , |
| • <i>L. grippityphosa</i> , | • <i>L. cynopteri</i> , |
| • <i>L. icterohaemorrhagiae</i> , | • <i>L. javanica</i> , |
| • <i>L. patoc</i> , | • <i>L. pomona</i> |
| • <i>L. pyrogenes</i> , | • <i>L. ratnapura</i> |
| • <i>L. valbuzzi</i> , | |



Microscopic Agglutination Test



IVRI maintains 21 most prevalent serovars of *Leptospira* in India & routinely perform MAT on sera samples

- Gold Standard Test for detecting leptospirosis
- Highly sensitive when performed on paired sera (acute and convalescent) and is serovar/ serogroup specific

Limitations

- Can give false negative test result when sera is collected from a patient in the early stages of the disease or when the patient is infected with a serovar which is not included in the test
- Cannot differentiate between vaccinated and naturally infected animal sera

Sero prevalence in elephants at different captive facilities

11. Parasitic Diseases

Parasitic diseases in elephants encompass a wide range of helminths, protozoa, and ectoparasites. While rarely causing acute mortality in healthy adults, parasitic burdens contribute to debilitation, secondary infections, and increased susceptibility to other diseases — particularly in captive elephants on restricted diets or in poor body condition.

11.1 Helminth Parasites

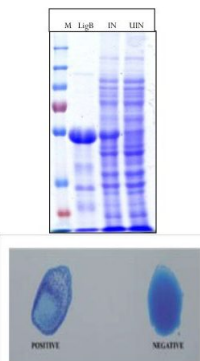
Parasite	Type	Location	Clinical Significance
Fasciola jacksoni	Trematode (Liver fluke)	Bile ducts / Liver	Fasciolosis: hepatomegaly, biliary fibrosis, emaciation; Fasciola eggs with bile in faeces
Pseudodiscus collinsi	Amphistome (rumen fluke)	Large intestine / Caecum	Heavy burdens → colitis; diarrhoea; seen on PM as reddish-brown flukes attached to mucosa
Strongylus spp. / Strongyles	Nematode	Anterior/posterior intestine	Colitis, protein loss; larvicidal migration causes hepatic tracking
Toxocara (in calves)	Nematode	Small intestine	Visceral larva migrans; pot-belly; poor growth in calves
Oxyuris (pinworm)	Nematode	Rectum	Tail rubbing; anal irritation; common in captive elephants

Figure 11.1 — Helminth parasites: *Pseudodiscus collinsi* (amphistome), GI nematodes (Strongyle), and *Fasciola jacksoni* eggs and flukes

Latex Agglutination Test

Purified LigB, LipL32 and LipL41 proteins are routinely employed in Latex Agglutination Test to screen field sera samples for detecting Leptospirosis and the results are compared with MAT

- LAT employs only OMP of Leptospira not the whole pathogen
- Serves as a pen site diagnostic test
- Able to detect infection caused by any pathogenic leptospiral serovars

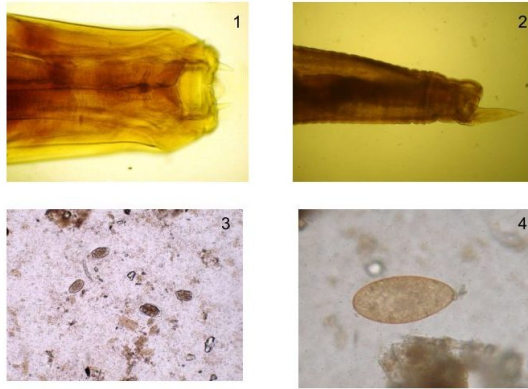



11.2 Protozoal Parasites

Protozoal Parasites

- **Balantidium coli/suis:** Large ciliated protozoan; infects colon; causes yellow diarrhoea; photomicrograph shows cysts and trophozoites; Treatment: Metronidazole 15–25 mg/kg BID for 5 days + Omeprazole
- **Haemoparasites:** Trypanosoma spp. found in blood smears; transmitted by biting flies; associated with emaciation, anaemia, lymphadenopathy; treated with Diminazene aceturate
- **Sarcocystis:** Intramuscular cysts; incidental finding at necropsy in most cases
- **Amoeba:** Entamoeba spp. occasionally detected in faecal smears; significance unclear in elephants

Figure 11.2 — Protozoal parasites: *Balantidium* cyst microscopy, yellow diarrhoea case, blood smear with haemoparasite



Anterior (Fig.1) and posterior end (Fig.2) of GI nematode (Strongyle worm) of Elephant;
Eggs of GI nematodes (Fig.3); Egg of *Fasciola jacksoni* (Fig. 4)



Fluke: *Fasciola*



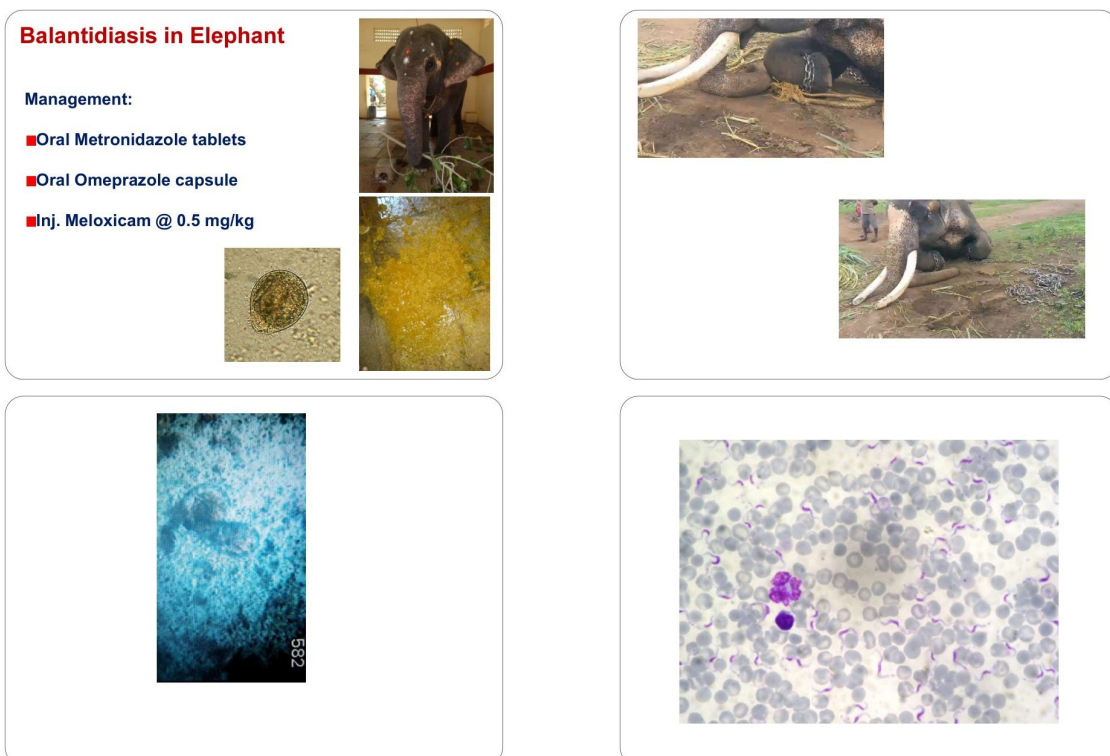
12. Surgical and Non-Infectious Conditions

Beyond infectious disease, field and zoo veterinarians frequently encounter traumatic, degenerative and management-related conditions. These require hands-on surgical skills and a good knowledge of elephant anatomy:

Common Surgical and Management Conditions

- **Sinus wound / abscess:** Common in captive elephants; secondary to bullhook injuries, trauma or tusk injuries; requires debridement, irrigation with povidone-iodine, and systemic antibiotics; may need surgical debridement under chemical restraint (Etorphine / Medetomidine)
- **Corneal opacity / eye conditions:** Corneal ulcers from foreign bodies, trauma or infection; treatment: topical antibiotic + atropine; assess retinal integrity; ophthalmoscopy under chemical restraint
- **Foot problems:** Extremely common in captive elephants — overgrown nails, foot pad cracks, sole abscesses; caused by hard substrate, poor hygiene, inadequate exercise; treatment: trimming, foot soaks (copper sulphate/betadine), systemic antibiotics
- **Dystocia:** Rare but life-threatening; calf mal-presentation; requires controlled delivery under sedation; pre-positioning of delivery assistance equipment recommended at breeding herds
- **Colic / Impaction:** Abdominal pain; anorexia; reduced faecal output; treat with liquid paraffin per os, fluids, Buscopan; severe cases may require flank examination under sedation

Figure 12.1 — Surgical cases: sinus wound treatment (Dudhwa NP), corneal opacity, foot nail overgrowth and pad cracks



Left: Sinus wound irrigation and debridement in elephant Roopakali (Dudhwa NP). Right: Corneal opacity assessment; overgrown foot nails; sole cracks in captive elephant.

13. Emerging Threats and Mass Mortality Events

Wildlife disease ecology is constantly evolving. RNA viruses, climate-driven pathogen range shifts, and novel toxicological threats pose growing risks to elephant populations. Veterinarians and forest officers must remain alert to unusual mortality clusters that may signal emerging pathogen activity.

Emerging Threats to Asian Elephant Health

- **Cyanobacteria / Algal bloom toxins:** Botswana 2020 — largest documented elephant mass mortality event; 350 elephants died around waterholes; cyanobacterial toxins (suspected nodularin/microcystin or BMAA) implicated; elephants showed neurological signs before collapse
- **Monkeypox (now Mpox):** Documented in captive elephants from Africa; can spillover to keepers; vesicular skin lesions; zoonotic potential elevated since smallpox vaccination stopped
- **Novel RNA viruses:** Bat-origin coronaviruses (SARS-CoV-2 antibodies detected in some captive elephants in Belgium); Paramyxoviruses; Nipah (bats → elephants theoretically possible in Kerala)
- **Climate change:** Shifts in mosquito vectors → altered arbovirus range; changes in grass/crop phenology affecting mycotoxin risk; increased heat stress mortality events
- **Antibiotic resistance:** Elephants in close proximity to human settlements accumulate resistant bacteria; antibiotic sensitivity testing mandatory before treatment

Figure 13.1 — Botswana 2020: mass elephant mortality around waterhole (cyanobacteria)

Successful treatment of sinus wound in Asiatic elephant at Dhudhwa National Park

Elephant Roopakali

ABST - Amikacin, Streptomycin and colistin



Corneal opacity and cataract



Over grown nails with cracks

Cracks in the foot pad

14. Field Practice Principles and Preparedness

Effective disease diagnosis and management in the field requires structured protocols, appropriate equipment, and a systematic approach. The following principles should guide field veterinarians at any elephant casualty or sick elephant call:

14.1 Antibiotic Use and Sensitivity Testing

- Always culture-based treatment when possible — empirical antibiotics only in emergencies
- Collect swabs/tissue/blood for culture BEFORE starting antibiotics
- Disk diffusion / Kirby-Bauer method for sensitivity testing (readily available at district labs)
- Avoid broad-spectrum antibiotic overuse — accelerates resistance in zoonotic bacteria
- Document all drug use: drug name, dose, route, duration, outcome — essential for case analysis

14.2 Documentation and Photography Protocol

- **Photograph before touching:** Establish GPS location; photograph the scene and carcass position before examination
- **Video documentation:** Record gross PM findings continuously; voice narrate observations
- **Systematic PM photography:** Each body cavity; each organ; all lesions with scale bar
- **Labelling:** Sample labels in waterproof ink; photograph each labelled sample with ID visible
- **Chain of custody:** Log each sample handed over; signatures required
- **Necropsy report:** Submit within 48 hours using standard DFO/CWL format

14.3 Essential Field Kit

Category	Items
Personal Protective Equipment	Nitrile gloves (×3 pairs/person), N95 mask, splash goggles, Tyvek suit, boot covers, biohazard waste bags
Sample Collection	EDTA tubes, plain red-top tubes, formalin pots (10%), sterile swabs, viral transport media, bone marrow aspiration needle, labelled zip-lock bags
Instruments	Scalpel (large handle + #22 blades), bone saw or hacksaw, forceps, scissors, ruler, weighing scale
Diagnostics	Portable glucometer, haemoglobin meter, glass slides, Wright/Giemsa stain, Seller's stain, darkfield microscopy (if available)

Communication	GPS unit, camera with macro lens, waterproof field notebook, contact list (IVRI, lab, WW office, PCCF)
Cold Chain	Ice packs, insulated cool box, cold chain labels

15. Conclusion and Key Take-Aways

The field of elephant health medicine in India is rapidly evolving, driven by increasing human–elephant interface, expanding captive elephant populations, and the emergence of novel pathogens. This chapter has presented the major infectious, parasitic, and management-related diseases of Asian elephants, with emphasis on field-relevant diagnostic and treatment protocols.

Key Take-Aways for Field Veterinarians

1. **EEHV** is the highest-priority disease for calves and young elephants — famciclovir must be pre-positioned; early clinical recognition saves lives.
2. **Anthrax** requires zero-necropsy protocol in the field — burn all material; never bury.
3. **Tuberculosis** is a chronic silent killer and a two-way zoonotic risk — mandatory surveillance of working elephants and their handlers.
4. **Rabies** — brain is the only diagnostic sample; full PPE mandatory; post-exposure prophylaxis for exposed personnel.
5. **HS and Clostridial** — bone marrow is the gold-standard postmortem sample; collect before lysis.
6. **Parasites** — regular faecal examination and strategic deworming are the cornerstone of captive elephant health maintenance.
7. **Documentation** — systematic photography, GPS recording and rapid lab submission are as important as the clinical intervention itself.
8. **Preparedness** — every elephant camp and zoo should maintain a preparedness kit, an emergency contact tree, and a supply of pre-positioned drugs (famciclovir, anthrax vaccine, HS vaccine, broad-spectrum antibiotics).

Acknowledgements

- **Resource Person:** Dr. M. Karikalan, MVSc, PhD — Assistant Professor, Dept. of Wildlife Science, Veterinary College & Research Institute (VBSC), Namakkal, Tamil Nadu
- **Institutions acknowledged:** Centre for Wildlife Studies (CWL), ICAR-IVRI, Wildlife Institute of India (WII), Nandankanan Zoological Park, MTR Nilgiris, ERRC Surajpur (Chhattisgarh), Dudhwa National Park
- **Workshop organisers:** Chhattisgarh Forest Department; Project Elephant Division, MoEF&CC; PCCF (Wildlife) Chhattisgarh State
- **Sponsors & Partners:** Central Zoo Authority (CZA), MoEF&CC;, Dr Parag Nigam (WII)

Figure 15.1 — Acknowledgements page from the source presentation



Boswana-Cyanobacteria (HCN)

Acknowledgment

1. CZA, MoEF & CC
2. Project Elephant, MoEF & CC
3. Director, ICAR-IVRI
4. Incharge, Centre for Wildlife, ICAR-IVRI
5. Dr Parag Nigam, WII
6. State Forest Department, Chhattisgarh(Team Raigarh)
7. Wildlife Veterinary Officers

Thank you



CHAPTER 4

Collection of Biological Materials

for Wildlife Disease Diagnosis

Dr. M. Karikalan, Senior Scientist (Veterinary Pathology) · Centre for Wildlife Conservation, Management & Disease Surveillance ·

Workshop on Essentials for Mortality Investigation of Asian Elephant · Chhattisgarh Forest Department · June 2026

1. Why We Collect Biological Materials — The Diagnostic Pipeline

Wildlife disease diagnosis works through several layers of investigation, each requiring specific biological materials collected at the right time, with the right preservation method, and dispatched to the appropriate laboratory. Understanding this diagnostic pipeline is essential before selecting which samples to collect.

The diagnostic pipeline encompasses:

- **Clinical observations** of the live or recently dead animal.
- **General health parameters** — invasive (haemato-biochemical profile) and non-invasive (scat/dung analysis).
- **Collection of specific biological materials** from diseased or dead animals.
- **Post-mortem (necropsy) examination** — gross lesions, histopathology (including histochemistry and immuno-histochemistry), microbiology (microscopy, culture, biochemical reactions, serological ID, molecular biology / PCR), toxicology, and parasitology.

How Disease Shows Up: Gene → Gross

Lesions develop in a sequence from the molecular level outward. Knowing this hierarchy tells you which diagnostic test detects what — and therefore which sample to collect and preserve.

Level	What Changes	How It Is Detected
1. Gene / Molecular	DNA mutation, epigenetic & gene-expression changes, abnormal protein synthesis	Not visible by light microscopy — requires PCR / molecular methods
2. Ultrastructural	Mitochondrial swelling, RER dilatation, ribosome detachment, membrane blebbing, chromatin clumping	Electron microscopy only
3. Light-Microscopic	Cellular swelling, fatty change, necrosis, apoptosis, inflammation, fibrosis, hyperplasia / dysplasia / neoplasia	Routine histopathology (H&E; stain)
4. Gross (Macroscopic)	Organ enlargement, tumour mass, ulcer, infarct, haemorrhage, abscess, tissue necrosis	Visible to the naked eye at necropsy

Key Concept: Lesion Hierarchy

Molecular lesion → Ultrastructural lesion → Microscopic lesion → Gross lesion

Most field investigations start at the gross level — but PCR can detect molecular lesions before any gross changes appear. Collect samples for BOTH.

2. Clinical Examination & the Samples It Generates

Common clinical workups include physical examination, haematology, serum biochemistry, urine examination, faecal examination, biopsy/FNAC, lateral flow assay, X-ray, and ultrasonography. For each workup, specific biological materials must be collected correctly to obtain valid results.

Clinical samples commonly collected from elephants:

- **Blood in anticoagulant vials** — EDTA or heparin (4–5 mL)
- **Serum** in a plain tube (allow to clot, then separate)
- **Faecal / rectal swabs or dung**
- **Ocular / nasal / oral swabs** — with or without Viral Transport Medium (VTM)
- **Trunk wash fluids** — critical for TB and EEHV diagnosis in elephants
- **Skin scrapings / hair follicles**
- **Any lesions** — abscess / wound discharge / tissue biopsy

Planning Rule

Always plan around two questions before collecting any sample:

- (1) Which laboratory test is intended for this sample?
- (2) What biosafety precautions and preservation method are required?

3. Master Sample-Collection Table

For each sample type: why you collect it, how to preserve / collect it, and how to store and transport it to the laboratory.

Sample	Purpose	Preservative / Collection Method	Storage & Transport
Blood	Infections, anaemia, general health	5–6 mL in EDTA vial	4–8 °C (refrigerated)
Blood smear	Blood parasites, cell abnormalities	Methanol-fixed blood smear on clean glass slide	Wrap in dry soft paper; room temperature
Plasma	Toxicity & drug-metabolite analysis	5–6 mL in heparin vial; transfer plasma to sterile tubes	–20 °C / –80 °C, or dry ice
Serum	Infections (antigen & antibody), health parameters, hormones	5–6 mL in plain serum vial; after clotting transfer serum to sterile tubes	4–8 °C short term; –20/–80 °C long term; dry ice for transport
Dung (faeces)	Parasites	20 g in sterile vial / zip-lock; worms in 10% formalin or 70% ethanol	4–8 °C; room temperature
Dung	Digestive disorders	100–200 g in sterile vial	4–8 °C
Dung	Hormone analysis	100–200 g in sterile vial / zip-lock	4–8 °C
Urine	Kidney function, UTIs	20–30 mL aseptically in sterile vial	4–8 °C
Urine	Bacterial infections (e.g. <i>Leptospira</i>)	Drops into a tube with specific bacterial media (e.g. <i>Leptospira</i> media)	4–8 °C
Saliva	Bacterial & viral infections	1–2 mL in sterile tubes or sterile swabs	4–8 °C (bacteria); –20/–80 °C (viruses)

Sample	Purpose	Preservative / Collection Method	Storage & Transport
Nasal discharge / trunk wash	Respiratory disease, esp. TB & EEHV	20–30 mL trunk wash using normal saline in sterile vial	–20/–80 °C for EEHV & TB; 4–8 °C for bacteria
Ocular discharge	Eye infections	Sterile tube/swab, ± specific media	4–8 °C (bacteria); –20/–80 °C (viruses)
Biopsy tissue / skin swabs	Skin & tissue diseases	Swabs aseptically ± transport media; tissues in 10% buffered formalin	4–8 °C for swabs; room temperature for formalin-fixed tissue

4. Examination-Specific Protocols

4a. Histopathological Examination

- Collect tissue **soon after death**; fix **immediately** in **10% buffered formalin**.
- Take several pieces — from the **centre** and from the **edge of the lesion** at the junction with healthy tissue (especially critical for large lesions).
- Tissue thickness: **0.5–1 cm**; formalin volume must be at least **10–15× the volume of the tissue** (minimum ratio: 10× the tissue amount).
- Use a **sharp instrument** (scalpel, knife, or razor) for a clean, crush-free cut.
- Always include **lesion tissue plus adjacent healthy tissue** in the same block.

Preparing 10% Neutral Buffered Formalin (NBF)

Commercial (saturated) formalin $\approx$ 37–40% formaldehyde. **10% NBF $\approx$ 4% formaldehyde** — this is the standard histopathology fixative. Tissue : fixative ratio = **1 : 10** at minimum (e.g. 10 g tissue in 100 mL of 10% NBF).

Ingredient	Quantity
Saturated formalin (37–40% formaldehyde)	100 mL
Distilled water	900 mL
Sodium phosphate monobasic ($\text{NaH}_2\text{PO}_4 \cdot \text{H}_2\text{O}$)	4.0 g
Sodium phosphate dibasic (Na_2HPO_4)	6.5 g
TOTAL	1,000 mL (1 litre)

Method: Dissolve both phosphate salts fully in ~800 mL distilled water → add 100 mL saturated formalin → top up to 1,000 mL → mix → label (name, concentration, date, preparer).

Simpler unbuffered version: 100 mL saturated formalin + 900 mL distilled water ($\approx$ 4% formaldehyde) — adequate when buffered NBF is unavailable in the field.

Preparation Notes

Use clean glass or plastic containers.

Dissolve phosphate salts **BEFORE** adding formalin.

Store at room temperature, tightly capped; label with name, concentration, and date.

4b. Bacteriological & Virological Examination

- Keep everything **free from contamination**; use **sterile instruments** (scalpels, scissors, syringes, bottles, slides, swabs, polythene bags).
- **Disinfect skin** before the primary incision to prevent surface contamination.
- Sear the organ surface (e.g. heart) with a **hot iron spatula** before sampling heart blood — to sterilise the surface and eliminate external contamination.
- Sample **abnormal areas** near the **edge of affected tissue** where live organisms are most likely still viable.
- **Appropriate samples**: whole blood, serum, CSF, pus, abscess/nodule areas, intestinal contents (within a tied loop), smears of pus or infected tissue.

For virology: use 5–10× volume of sterile 50% buffered glycerol, chilled PBS (pH 7.4), or Viral Transport Medium (VTM). Other acceptable preservatives: RNA Later, 70% ethanol (for PCR-based detection).

Transport: bacteria at 4 °C; viruses at 4 °C or on dry ice / frozen.

Heart-blood smears: heat-fix for bacteria only; fix in absolute methanol for haemoparasites.

4c. Parasitological Examination

- **Blood smears** on clean glass slides — fix in **absolute methanol**.
- **Skin scrapings** → preserve in **10% KOH**.
- **2 g faeces** → **70% ethyl alcohol** for molecular work; **formalin** for morphology-based identification.
- **Larvae, worms, maggots** → **70% alcohol** or **10% formalin** (room temperature).
- **Protozoan diseases** → lymph nodes, blood, and faecal sample in **5% formalin**.

4d. Toxicological Examination

- Collect **≥ 100 g of stomach + intestinal contents** and **≥ 100 g of liver + kidney** in a glass container with saturated salt solution* or ice.
- Also collect **environmental samples** — baits, feed/fodder, water sources near the mortality site.
- **Pack each sample type separately** in an ice box; do not mix organ samples with GI contents or environmental samples.
- All containers must be **serially numbered, properly labelled, and kept under safe custody** — chain of custody is legally critical in forensic cases.

■ Important — Qualitative vs. Quantitative Toxicology

*Saturated salt solution is NOT used when toxicants must be quantified.

For quantitative toxicology: use sterile containers / zip-locks and deep-freeze at –20 °C.

Toxicology sample guide — ante-mortem, post-mortem, and environmental:

Sample	Amount	Notes
ANTE-MORTEM		
Whole blood	5–10 mL	EDTA or heparin vial
Serum	5–10 mL	Remove from clot; use trace-element tubes for zinc assays
Urine	25–50 mL	Plastic screw-capped tube
GI contents	≥ 100 g each	Representative stomach/rumen contents; faeces
Hair	1–2 g	Clean zip-lock pouch
Milk	30 mL	Sterile collection tube

Sample	Amount	Notes
CSF	1–2 mL	Sterile tube; refrigerate immediately
POST-MORTEM		
Blood/serum/urine (from heart)	As ante-mortem	As per ante-mortem guidelines above
Liver	100–250 g	Plastic container; –20 °C
Kidney	100–250 g	Plastic container; –20 °C
Brain	One half	Sagittal cut; plastic; –20 °C
Fat	100 g	Foil inside plastic bag; –20 °C
GI contents	≥ 100 g each	Representative samples; PACKAGE SEPARATELY from organs
Ocular fluid / eyeball	As much as possible	Aqueous fluid preferred; or whole eyeball
Bone	≥ 100 g	For heavy metal analysis
Spleen	100 g	As for liver
Lung	100 g	As for liver
Hair or skin	—	If dermal exposure suspected
ENVIRONMENTAL		
Bait / source material	200 mL or g	Clean jar / whirl-pak; label precisely
Feed	≥ 500 g	Composite sample per lot/batch
Plants	Whole or representative	Fresh-pressed-dried, or frozen
Water	≥ 1 L	Clean glass jar; plastic lid for metals analysis

4e. Pregnancy Diagnosis & Deficiency Diseases

Pregnancy diagnosis in carnivores and elephants: collect **100–150 g of dung** plus serum samples for hormone analysis.

Deficiency diseases: collect **serum samples** for macro- and micro-nutrient assessment.

5. Elephant-Specific Collection Notes

- **Blood:** drawn from an **ear vein**; make blood smears immediately and fix in 70% / absolute methanol.
- **Trunk wash fluid:** the elephant draws water into the trunk, washes the inside, and the used water is expelled into a clean container. This is the critical sample for **TB and EEHV** diagnosis and should be stored at –20/–80 °C.
- **Rectal and oral swabs:** collected with sterile swab sticks, placed in zip-lock pouches (± VTM); store in RNA storage tubes or RNA Later as needed for molecular diagnosis of EEHV.

Elephant Priority Sample: Trunk Wash

Trunk wash is the MOST IMPORTANT sample for EEHV and TB diagnosis in elephants.
Freeze at –80 °C (or on dry ice for transport) if EEHV is suspected.
Do not refrigerate only — EEHV degrades rapidly at 4 °C.

6. Packing & Transport

Preferred Method — Fresh & Frozen Tissue

- a. Place **absorbent material** (cat litter or paper towel) in the outer bag to absorb spills.
- b. Tissues → individual, sterile, **labelled** plastic bottles or whirl-type bags.
- c. Containers + **frozen refrigerant packs** → heavy plastic bag labelled with contents.
- d. Bag → **insulated cooler** for shipment.
- e. **Specimen submission sheet + all documents** → envelope → plastic bag → taped to the **outside** of the package.

10-Step Packing Protocol (Zoonotic / General Samples)

1. Collect preferred samples with **all biosafety precautions** (gloves, mask, PPE).
2. **Label** specimens (animal name/species, age, gender, specimen ID, date, site).
3. **Parafilm-seal** the neck of all vials and containers.
4. Cover vials with **absorbent material (cotton)**.
5. Put tubes/vials in a **zip-lock pouch** and seal.
6. Place that pouch inside a **plastic container or another zip-lock** and seal tightly.
7. Put the container in a **thermocool box surrounded by hard-frozen gel packs**.
8. Put the **history form** in a separate zip-lock pouch inside the thermocol box.
9. **Close and seal** the thermocol box with adhesive tape.
10. Write the **address of the testing laboratory** clearly on top of the box.

Do's and Don'ts

Do ✓	Don't ✗
Use clean, properly labelled, leak-proof containers	Submit organs wrapped in plain plastic wrap
Use adequate fixative (10× formalin for histopath)	Use dirty or leaking jars / bottles
Parafilm-seal vial necks before packing	Under-fix tissue (too little formalin or too thick)
Maintain cold chain — refrigerate bacteria; freeze viruses	Ship a whole head or carcass loose in a leaking, poorly packed box
Include a complete history form and transport certificate	Mix samples from different organ types in the same container
Serial-number and record all forensic/toxicology samples	Break chain of custody for forensic samples

Transport Certificate

Include a signed certificate with every shipment stating that the samples are **non-corrosive, non-inflammable, and non-infectious**. The certificate must name the sender, species, sample type, origin, destination laboratory, and preservation method (ice gel packs / dry ice).

Example Certificate Wording

Sample wording: "Certified that samples of an Asian elephant (blood and trunk secretions) from [site], [date], are being transported by [scientist name] to ICAR-IVRI, Bareilly, for health examination; securely packed in sterile containers with frozen gel packs."

7. Essentials for Biological Sampling — Quick Master Reference

Diagnostic Activity	Specimen	Preservative / Container	Key Notes
Haematology	Whole blood	EDTA/heparin vial; ice pack 4–8 °C	Gently rotate to mix; do not keep at room temp for long
Serum biochemistry	Serum	Serum tube ± clot activator/gel; 4–8 °C or freeze	Handle gently; AVOID haemolysis
Histopathology	Tissues — liver, lung, kidney, spleen, heart, stomach, intestine, brain, lymph node, tumours (normal + abnormal)	10% NBF; leak-proof container; room temperature	0.5–1 cm thick; formalin:tissue ratio ≥ 10:1; include lesion + healthy margin
Bacteriology	Organs, ocular/rectal/naso- & oro-pharyngeal swabs, heart blood, exudates, intestinal loops, urine	Refrigerate 4–8 °C; nutrient broth / peptone water / PBS	Avoid contamination; sample type varies by suspected disease
Virology	Organs (small pieces), swabs, exudates, trunk wash, body fluids	VTM or PBS (pH 7.4); 50% glycerol saline	Refrigerate or dry ice; varies by disease
Parasitology — endoparasites	Nematodes, trematodes, cestodes	70% alcohol or 5–10% formalin	Room temperature
Parasitology — ectoparasites	External parasites (ticks, mites, flies)	10% KOH or 70% alcohol	Room temperature
Parasitology — blood	Blood parasites	Methanol-fixed blood smear	Glass slide at room temperature
Toxicology	~100 g organs, GI contents, blood, fat, water, fodder, feed	Qualitative: saturated salt solution. Quantitative: sterile containers; deep freeze –20 °C	Refrigerate for qualitative; freeze for quantitative; separate containers
Pregnancy diagnosis	Serum, scat/dung (100–200 g)	Plastic container / zip-lock; 4–8 °C	Refrigerate
Genetic / species ID	Tissues, blood, hair follicles, dung, bone marrow	Blood in EDTA; hair in pouches; tissue in 70% ethanol or dry ice	Refrigerate blood; freeze tissue/marrow; hair at room temp in dry pouch

8. Where to Send Samples

Services Available at ICAR-IVRI, Bareilly

- PCR diagnosis of important infectious diseases (EEHV, TB, Anthrax, FMD, Rabies, etc.)
- Bacterial isolation & antibiotic sensitivity testing (ABST)
- Toxicological investigations (qualitative and quantitative)
- Parasitology — endo-, ecto-, and haemoprotozoan parasites
- Pregnancy diagnosis via dung samples
- Necropsy, histopathology, and clinical pathology (blood, urine)
- Serodiagnosis of TB, Leptospirosis, and other infections

ICAR-IVRI Contact

Contact: In-charge, Centre for Wildlife

ICAR-IVRI, Izatnagar, Bareilly — cwlincharge@gmail.com

National Referral Laboratories by Specialty

Specialty	Recommended Laboratory
General wildlife / referral	National Referral Centre on Wildlife Health / Centre for Wildlife, ICAR-IVRI, Izatnagar
Exotic viral diseases (e.g. Avian Influenza)	ICAR-NIHSAD, Bhopal
Foot and Mouth Disease (FMD)	ICAR-Directorate on FMD, Bhubaneswar; IVRI Bengaluru
Toxicology	CSIR-Indian Institute of Toxicology Research (IITR), Lucknow
Forensic & molecular identification	Wildlife Institute of India (WII), Dehradun; CSIR-CCMB, Hyderabad
Poisoning incl. NSAIDs & avian forensic cases	SACON, Coimbatore (southern campus of WII)
Regional diagnostic labs (DADF)	Bangalore (KVAFSU Hebbal), Jalandhar, Pune (Aundh), Guwahati (Khanapara, NE region)
State veterinary universities / specialized centres	TANUVAS Chennai; KVASU Pookode Wayanad; OUAT Bhubaneswar; School of Wildlife Forensic & Health, Jabalpur (NDVSU); ICAR-NIVEDI Bengaluru

Key Takeaways**Chapter Summary — Biological Sample Collection**

1. Match the sample, preservative, and cold-chain to the intended test — the same organ goes into formalin for histopath, VTM for virology, and salt solution / freezer for toxicology.
2. Fix histology tissue IMMEDIATELY, thinly (0.5–1 cm), in 10× formalin volume, including both lesion and healthy margin.
3. Sterility and anti-contamination technique are critical for all microbiology samples.
4. Label, document, and cold-chain every sample; always include a transport certificate.
5. Toxicology and forensic samples require serial numbering and strict chain of custody.
6. Trunk wash is the priority sample for EEHV and TB in elephants — freeze immediately.

CHAPTER 5

Non-Infectious Diseases in Asian Elephants

Toxicology, Electrocution, Nutritional Disorders & Field Investigation Framework

Dr. Karikalan Mathesh, Senior Scientist · Centre for Wildlife Conservation, Management & Disease Surveillance · ICAR-IVRI, Bareilly

Workshop on Essentials for Mortality Investigation of Asian Elephant · Chhattisgarh Forest Department · June 2026

1. Background — Why Elephants Are Biologically Distinct

Asian elephants are physiologically unique in two seemingly contradictory respects: they are extraordinarily resistant to cancer, yet highly vulnerable to poisoning. Understanding both traits is essential for interpreting necropsy findings and identifying non-infectious causes of mortality.

Cancer Resistance — Peto's Paradox

Despite enormous body size and a 60–70 year lifespan (far more cells and far more time for mutations than in small animals), elephants have a **remarkably low incidence of cancer**. The key mechanism:

- **Multiple TP53 copies** — elephants carry **20+ copies** of the TP53 tumour-suppressor gene ("guardian of the genome"), compared with just **1 copy in humans**.
- **Enhanced DNA-damage detection** — more TP53 enables faster, more accurate identification of cellular DNA errors.
- **Stronger apoptotic response** — damaged cells are eliminated by programmed self-destruction before they can proliferate into tumours.

High Susceptibility to Toxicity

The same large body that should theoretically raise cancer risk instead makes elephants **highly prone to poisoning**. Several converging factors explain this:

Risk Factor	Mechanism / Consequence
Dysfunctional CYP genes	Cytochrome P450 (CYP) genes are dysfunctional → limited ability to biotransform and detoxify ingested toxins
Massive daily intake	Adults consume 100–300 kg of vegetation per day; toxic plants, contaminated fodder, pesticides, and mycotoxins are ingested in large amounts
High water consumption	100–200 L of water per day — polluted sources expose them to heavy metals, pesticides, cyanobacterial and industrial toxins
Hindgut fermenters	Sudden diet changes or spoiled feed disrupt gut microbiota and increase endogenous toxin production
Non-selective feeding	Explore environment with the trunk; may consume unfamiliar plants, crops, garbage, or treated material without discrimination
Drug sensitivity	Extremely sensitive to certain veterinary drugs and anaesthetics; incorrect dosing can cause respiratory depression, cardiac failure, or death
Pollutant accumulation	Long lifespan (60–70 yrs) allows progressive build-up of heavy metals (lead, mercury, arsenic) and persistent organic pollutants

2. Case Study — Mass Mortality at Bandhavgarh NP (Oct 2024)

History & Field Observations

Incident: Sudden death of **10 elephants from a herd of 13** on **31 October 2024**. During forest-department patrolling, several elephants were found recumbent with weak respiration; some were already dead.

Location: Salkhania beats of Pataur and Khitauli Ranges, Bandhavgarh National Park (Tiger Reserve), Madhya Pradesh.

- **Heavy rainfall** during October 2024.
- **Kodo millet crops were unharvested** near forest boundaries — a critical epidemiological clue linking the elephants to fungal-contaminated grain.

Clinical Signs & Gross Necropsy

Clinical progression: respiratory distress → ataxia → weakness → incoordination → recumbency → rapid collapse.

Organ / Site	Gross Finding
Lungs	Congestion and oedema
Liver	Enlarged, congested and swollen
Kidneys	Congested
Body cavities	Serosanguinous (blood-tinged) fluid accumulation
Multiple organs	Congestion and haemorrhages throughout
Heart	Pale necrotic foci on the myocardium

Histopathology

Tissue	Histopathological Finding
Heart	Acute myocardopathy — degeneration, vacuolation and necrosis of cardiac myofibres
Liver	Hepatocellular degeneration — centrilobular necrosis, fatty change
Kidney	Renal tubular necrosis with epithelial degeneration
General	Congestion, haemorrhage and oedema throughout multiple organs

Sample Collection & Investigation Pathway

Sample	Preservation	Purpose
Tissues in 10% NBF	Formalin fixation	Histopathology
Tissues on ice + heart blood	Chilled	Disease screening (EEHV, EMCV, HS, Clostridium)
Stomach contents, liver, kidney, heart, fat	—	Toxicology
Hair follicles / nails	—	Toxicology (chronic exposure indicators)
Environmental samples (water, feed, fodder, grain)	—	Toxicology

Toxicological methods: conventional tests → thin-layer chromatography (TLC) → **liquid chromatography–mass spectrometry (LC-MS)**.

Toxicology Results

Negative for: HCN, nitrate-nitrite, heavy metals, organophosphates, organochlorines, pyrethroids, carbamates.

Negative for infectious causes: EEHV, pasteurellosis, other mycotoxins.

Positive — Cyclopiazonic acid (CPA): detected in all pooled samples at **>100 ppb** (Ref. CAD-877/2024-25, dated 05-11-2024). Results indicated the elephants had consumed large quantities of Kodo plant/grains.

LC-MS Confirmation (ICRISAT, Hyderabad)

23 samples received 6 November 2024. High CPA detected in stomach contents, tissues, and millet samples:

Sample Description	CPA (ppb)
Kodo grain with straw (range)	~1,567 – 12,874
K-2 (Kodo grain + straw)	12,874
K-5 (Kodo grain + straw)	10,667
Vomit sample (C3, Kodo millet)	4,486

Final Diagnosis & Pathophysiology

DIAGNOSIS: Toxicosis — Kodo millet (cyclopiazonic acid) poisoning.

Pathophysiology: CPA caused acute cardiac injury (myocardiopathy) → cardiac arrest / heart failure → passive venous congestion → hypoxia → degeneration of parenchymatous organs (liver, kidney, spleen). Findings correlate fully with necropsy gross lesions and laboratory results.

3. Cyclopiazonic Acid (CPA) — Key Facts

Source, Chemistry & Toxicity

- **Source plant:** *Paspalum scrobiculatum* (kodo / koda millet). Grown mainly in Nepal; also India, Philippines, Indonesia, Vietnam, Thailand. Originated in West Africa.
- **CPA is a mycotoxin** — a secondary metabolite of fungi *Aspergillus* and *Penicillium*. Kodo seed is particularly infected by *A. flavus* and *A. tamarii*.
- **Chemistry:** α -cyclopiazonic acid, an indole-tetramic acid; molar mass $\approx$ 336.39 g/mol; PubChem CID 54682463.
- **Key hazard:** Ripe kodo can become intermittently poisonous ("Kiruku Varagu") without any visible change — fitness for consumption cannot be judged by appearance alone.
- **First human report** linking CPA to kodua poisoning: Rao & Husain (1985), *Mycopathologia*.

Historical Precedent — 1933

R.C. Morris, *J. Bombay Nat. Hist. Soc.* (1934) documented the death of **14 elephants** near Vannathiparai Reserve Forest (December 1933) from kodo millet ("Varagu") poisoning. The traditional antidote was **tamarind-water or buttermilk in large quantities**; one of three elephants given water and tamarind reportedly survived (unconfirmed). This demonstrates that Kodo poisoning of elephants is a long-recognised, recurring problem.

Prevention & Control

1. **Regulate agriculture** and enhance monitoring of fungal-infected cropping systems near forest boundaries.
2. **Community awareness** — engage farmers and livestock owners around the reserve.
3. **Prevent elephant access** to contaminated, unharvested millet fields.
4. **Scientific surveillance** and systematic monitoring of kodo crops.
5. **Inter-departmental coordination** between forest, agriculture, and revenue departments.

Advisory & One Health Angle

Correlate history with necropsy; survey and destroy fungal-infected Kodo residue; keep domestic animals and wildlife out of affected fields; conduct cropping/ambient studies; determine the LD50 of CPA in domestic and wild animals.
ONE HEALTH ALERT: Cattle in the area were also affected. Watch for symptoms in humans — Kodua poisoning is a recognised human mycotoxicosis.

4. Other Major Non-Infectious Causes of Elephant Mortality

A. Electrocution (Electrical Injury)

Electrocution is one of the most commonly reported non-infectious causes of elephant death in India, particularly from power lines and illegal electrified fences.

Mechanisms of damage:

- **Direct:** cell depolarisation, electroporation of cell membranes.
- **Indirect / thermal:** current generates heat → thermal burns and coagulative necrosis.
- **Secondary mechanical:** violent muscle contraction or cardiac arrest causes falls and crush injuries.
- Final common pathway: **cell death (necrosis) and organ failure.**

Comparison of Injury Patterns

Feature	Lightning	High Voltage (>1 kV)	Low Voltage (<1 kV)
Voltage	>30,000,000 V	>1,000 V	<1,000 V
Current	>200,000 A	<1,000 A	<240 A
Cardiac arrest	Asystole	Ventricular fibrillation	Ventricular fibrillation
Muscle contraction	Single (single jolt)	Depends on duration	Tetanic (sustained)
Burns	Rare, superficial	Common, deep	Usually superficial
Rhabdomyolysis	Uncommon	Very common	Common

Feature	Lightning	High Voltage (>1 kV)	Low Voltage (<1 kV)
Mortality	Very high	Moderate	Low

Current Magnitude & Physiological Effect

Reference thresholds (general): 3 mA — noticeable shock; 10 mA — muscular contractions; 30 mA — respiratory paralysis; 50 mA — heart paralysis (may be fatal); 100 mA — ventricular fibrillation (fatal); 4 A — heart paralysis (fatal); 5 A — tissue burning.

Tissue resistance determines the path and extent of injury. Nerves, blood, mucous membranes, and muscle have low resistance (higher injury). Tendons, fat, and bone have high resistance and are relatively spared.

Field Findings & Post-Mortem Clues

Of 10 reported electrocution cases in elephants: **6 from direct power-line contact (440–220 V)**, **4 from electrified fences**. Affected states: Uttar Pradesh (4), Tamil Nadu / Mudumalai NP (2), Uttarakhand (2), Chhattisgarh (2).

- **Very rapid decomposition** and quick or absent rigor mortis.
- **Lightning / burn marks** — charred skin, lacerations along the current path (e.g. trunk burns in the Dudhwa NP case).

B. Lightning Side-Potential (Herd Mortality)

A ground-strike lightning event creates a high-voltage point; current then spreads radially outward through the ground for **up to 100 m or more**. Elephants standing or lying on the ground become part of the electrical circuit. Current passes through the body between front and rear feet → **instant internal organ failure and death — without any direct lightning strike**. This mechanism explains multiple simultaneous deaths in closely grouped herds.

Field Clue — Lightning Side-Potential

High-risk conditions: open grasslands, proximity to water bodies, thunderstorms, and tightly grouped herds.

Documented in Assam (pre-monsoon / monsoon season).

Investigate GPS clustering of carcasses and weather/storm records immediately.

C. Infighting / Tusk Injuries

Penetrating tusk wounds from intraspecific combat are a recognised non-infectious cause of elephant mortality, particularly in adult males. Post-mortem examination reveals deep puncture wounds, haemorrhage, and secondary infection. (e.g. Najibabad Forest Division, Uttar Pradesh.)

D. Nutritional Deficiencies

Nutritional deficiency disease can be **primary** (inadequate dietary supply) or **secondary**, caused by one of several interfering mechanisms:

Mechanism of Secondary Deficiency	Examples / Notes
Interference with intake	Pain on eating, dental disorders, jaw disease

Mechanism of Secondary Deficiency	Examples / Notes
Interference with absorption	Chronic intestinal disease, parasitism, dysbiosis
Interference with storage / utilization	Liver disease impairing vitamin storage and metabolism
Increased excretion	Renal disease, diarrhoea, heavy parasite burden
Increased requirements	Pregnancy, lactation, rapid growth in calves
Anti-nutritional substances	Dietary inhibitors (e.g. phytates, oxalates) blocking nutrient uptake

Clinical examples in elephants:

- **Hypocalcaemic tetany and rickets** / bone disorders (e.g. "Yogi" at Rajaji NP, Dehradun).
- **Corneal opacity and cataract** — chronic micronutrient deficiency.
- **Overgrown, cracked nails and cracked foot pads** — chronic nutritional and management deficiency in captive or managed animals.

E. Advanced Decomposition / Skeletonization

Late-discovered carcasses (e.g. Amangarh Tiger Reserve, Bijnor) substantially limit diagnostic yield. Advanced autolysis and scavenging destroy soft tissue lesions, preclude sampling for microbiology and virology, and compromise toxicological analysis. This underscores the critical importance of **early detection, rapid reporting, and prompt necropsy**.

5. Field Investigation Framework — Exam-Ready Checklist

When facing a suspected wildlife disease outbreak or mass mortality event, the wildlife veterinarian must be able to address these six areas:

Area	Key Actions / Considerations
1. First Response	Secure the site. Ensure personnel safety and PPE. Notify field director and authorities. Prevent further access/exposure. Barricade and photograph the scene.
2. Initial Field Investigation	Record complete history: time of onset, clinical signs observed, GPS coordinates, environmental context (recent rainfall, crops nearby, water sources). Note scavenger activity and any other species affected (One Health).
3. Essential Requirements	PPE and biosafety kit; necropsy and sampling kit (scalpels, forceps, containers); cold chain (ice box, gel packs); fixatives (10% NBF vials); documentation (camera, field book); labelled containers; transport logistics and lab contacts.
4. Samples for the Laboratory	Fixed tissues for histopathology; chilled tissues + heart blood for disease screening; stomach contents, liver, kidney, heart, fat, hair/nails for toxicology; environmental water, feed, fodder, grain.
5. Differential Diagnoses	Infectious: EEHV, Pasteurellosis/HS, EMCV, clostridial disease. Toxic: mycotoxins/CPA, HCN, nitrate-nitrite, pesticides, heavy metals. Physical: electrocution, lightning side-potential, tusk injuries. Nutritional/metabolic deficiency.
6. Prevention & Control	Apply the five-point plan (regulate agriculture, community awareness, prevent access, scientific surveillance, inter-departmental coordination). Operate under a One Health framework linking wildlife, livestock, and human health.

6. Key Takeaways

Chapter Summary — Non-Infectious Diseases in Asian Elephants

1. Elephants resist CANCER (20+ TP53 copies) but are highly VULNERABLE TO TOXINS (dysfunctional CYP genes + massive daily intake of vegetation and water).
2. The Bandhavgarh 2024 die-off was toxicosis from cyclopiazonic acid (CPA) in fungal-infected, unharvested Kodo millet after heavy rains — confirmed by LC-MS.
3. Lesion pattern: acute myocardiopathy → circulatory failure → multi-organ hypoxic/toxic degeneration (liver, kidney, spleen).
4. Kodo millet poisoning is historically documented (since 1933) and has One Health relevance: cattle and humans in the same area are also at risk.
5. Other leading non-infectious causes in India: electrocution (power lines & fences), lightning side-potential (herd deaths), infighting, and nutritional / metabolic deficiencies.
6. Early detection, prompt necropsy, and complete sampling are essential — late-stage decomposition destroys diagnostic evidence.

CHAPTER 6

Non-Infectious Pathology in Asian Elephants

Poisoning, Electrocution, Reproductive & Metabolic Disorders

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Workshop on Essentials for Mortality Investigation of Asian Elephant · Chhattisgarh Forest Department · June 2026

1. Elephant Poisoning in India — Overview

Poisoning is one of the most severe threats to **pachyderms** — the large, thick-skinned mammals including elephants, rhinoceroses, and hippopotamuses. In India, unnatural causes such as electrocution and poisoning claim hundreds of elephant lives every year. Incidents arise primarily from human–wildlife conflict and illegal poaching, and parallel patterns are seen in Africa.

Route	Examples & Context
Intentional	Illegal poaching; retaliatory killing in human–elephant conflict areas. Deliberate placement of poisoned bait or contaminated food near crop fields.
Accidental	Elephants consuming mycotoxin-infected crops (e.g. kodo millet); drinking from algae-bloom water bodies; ingesting toxic plants unknowingly.

2. Mycotoxins & Biological Toxins

A. Crop-Borne Mycotoxins (CPA / Kodo Millet)

Elephants raiding farm fields are highly susceptible to **mycotoxins** — secondary metabolites of moulds. The most clinically significant is **Cyclopiazonic Acid (CPA)**, produced by *Aspergillus* and *Penicillium* species infecting **kodo millet** crops.

Pathological effects: acute vascular damage, liver necrosis, and kidney necrosis, ultimately leading to multi-organ failure and death.

Bandhavgarh Case Reference

Key case — Bandhavgarh Tiger Reserve, MP: A herd of 13 elephants raided kodo millet crops; 10 elephants died. Forensic post-mortem and lab investigation confirmed accidental toxicity from fungal mycotoxins in infected crop.
(Speaker note: 2 similar cases personally seen/treated in KTR camp elephants, 1997–98.)

When Does Contamination Occur?

Heavy or unseasonal rainfall coinciding with grain maturation/harvest creates moist, warm conditions ideal for fungal colonisation. Unharvested grain in standing fields carries the highest risk.

Management & Prevention:

- Good agricultural practices** — correct spacing, timely harvesting, and minimising field duration after maturity.
- Biocontrol agents** — application of non-toxigenic fungal strains (*Aspergillus* spp.) to competitively exclude toxin-producing strains.

3. Post-harvest handling — thorough drying and airtight storage to prevent moisture-driven fungal re-growth.

B. Cyanotoxins / Cyanobacteria

Toxic blue-green algae (cyanobacteria) proliferate in stagnant, warm, nutrient-rich water bodies during **hot, dry seasons**. Dense blooms produce potent **neurotoxins and hepatotoxins** that can cause fatal mass die-offs — especially at elephant watering holes where large volumes of contaminated water are consumed rapidly.

C. Toxic Plants

Elephants generally avoid toxic vegetation through learned behaviour, but unfamiliar environments, food scarcity, or captivity can lead to accidental ingestion. Key examples:

Plant / Group	Toxic Mechanism & Effect	System Affected
Oleander (<i>Nerium oleander</i>)	Cardiac glycosides → fatal arrhythmia and cardiac arrest	Cardiac
Certain weeds / unfamiliar plants	Variable alkaloids, glycosides → neurological signs	Neurological
Calcium oxalate plants (e.g. <i>Dieffenbachia</i>, aroids)	Crystal penetration of oral/GI mucosa → intense swelling, irritation, and systemic distress if eaten in quantity	Mechanical + systemic

3. Hydrocyanic Acid (HCN) Poisoning

Source & Risk Factors

Cyanogenic glycosides in **jowar / sorghum (*Sorghum bicolor*)** are hydrolysed to release HCN (prussic acid). Risk is highest in:

- **Young, rapidly growing plants** — glycoside content peaks before maturity.
- **Stunted or stressed plants** — e.g. following drought then autumn rains.
- **Leaves and stalks** — contain far more glycoside than grain.
- Feeding elephants **large, unfamiliar quantities** of fresh jowar at once (especially when regular feed is unavailable).

Case Study — 5 Captive Elephants (Circus, Jabalpur)

Reference: Zoo Print, Vol. XXV, 6 June 2010.

Five circus elephants were fed large quantities of fresh jowar (unavailability of regular feed). They developed **restlessness and diarrhoea**; routine anti-diarrhoeal treatment failed and condition worsened. HCN poisoning was suspected and the following treatment was instituted:

Step	Action
Step 1	Stopped jowar feeding immediately.
Step 2	Dextrose saline (8–10 L) + 30 mL multivitamin injection (MVI) via ear vein.
Step 3	Sodium thiosulphate (50 g) dissolved in water, given orally as antidote.
Step 4	Treatment repeated at approximately 12-hour intervals.

Step	Action
Step 5 (critical cases)	Additional doses + cold dextrose saline + small ice pieces + Tab. Diadin per rectum.
Outcome	Recovery achieved; all animals resumed drinking and grazing.

Key Learning Point

Lesson: HCN poisoning here was accidental — caused by feeding large amounts of unfamiliar sorghum to elephants not accustomed to it, due to non-availability of regular food. Sodium thiosulphate is the key antidote (converts cyanide to thiocyanate).

4. Electrocution

Electrocution is death or severe injury caused by an electric shock. The body becomes part of an electrical circuit; high voltage or current disrupts the body's natural electrical systems, commonly triggering **cardiac arrest or severe electrical burns**.

Term	Definition
Electric Shock	Physical sensation or injury from electricity — muscle spasms, minor burns. NON-FATAL.
Electrocution	An electric shock that is FATAL. The term implies death as the outcome.

A documented case of elephant electrocution was recorded in November 2020.

5. Pyometra

Definition & Affected Population

Pyometra is a severe, life-threatening **uterine infection** marked by accumulation of pus and suppurative inflammation of the uterus. It occurs most commonly in **aging, nulliparous (never-given-birth) female elephants** and is linked to prolonged hormonal fluctuations and underlying reproductive issues.

Causes & Pathophysiology

Factor	Mechanism / Details
Hormonal changes	Fluctuating progesterone levels → abnormally thickened uterine lining (endometrial hyperplasia) creates a nutrient-rich environment for bacterial growth.
Bacterial infection	Secondary invasion by organisms such as <i>E. coli</i> or <i>Enterococcus faecium</i> → suppurative inflammation and pus accumulation.
Pre-existing conditions	Cystic endometrial hyperplasia or benign tumours (leiomyomas) in older captive females predispose to pyometra.

Treatment

- 1. Hormonal downregulation** — GnRH vaccines (e.g. Improvac) or Deslorelin implants to suppress reproductive cycles and reduce endometrial stimulation.
- 2. Aggressive antibiotic therapy** — broad-spectrum systemic antibiotics targeting specific isolated pathogens.

3. Uterine lavage — flushing and draining the uterus to remove accumulated pus.

4. Homeopathic adjunct protocol practised: Pyrogenium 1000 + Hepar Sulph 1000 + Secale 1000.

Surgical management: vaginal vestibulotomy has been performed in Asian elephant cases.

6. Hyperthermia

Why Elephants Are Prone to Overheating

Hyperthermia is a life-threatening condition arising from high environmental temperature, lack of access to water/shade, massive body volume, and the absence of conventional sweat glands. Three converging factors explain the heightened risk in elephants:

Risk Factor	Explanation
Square-Cube Law	As body volume increases, metabolic heat production scales faster than the relatively smaller skin surface area available to dissipate it. Large size = excellent heat retention, poor heat loss.
No sweat glands	Elephants lack traditional eccrine sweat glands; only pores between the toes are present. They depend on transepidermal water loss through the skin — which requires ready access to drinking water.
Continuous metabolic heat	Sustained walking or exertion in direct sun generates metabolic heat faster than passive cooling can remove it, triggering hyperthermia in the absence of active cooling behaviour.

Natural Cooling Mechanisms

- **Ear flapping** — the large, heavily vascularised pinnae act as thermal radiators. Rapid flapping increases convective heat loss from blood circulating through the ears, lowering overall body temperature.
- **Wallowing and water/mud spraying** — immersion or spraying with water provides powerful evaporative cooling; a residual mud coat shields the skin from solar radiation (natural sunscreen).
- **Saliva spraying** — self-spraying with trunk-collected saliva adds evaporative surface cooling.
- **Increased skin permeability** — the hide becomes more permeable during heat stress, allowing greater transepidermal evaporative cooling — but only if adequate drinking water is available.

Management of Hyperthermia

Intervention: Provide unlimited cool drinking water and shade immediately.

Apply cold water sprays to the skin, behind the ears, and on the feet.

Move the animal out of direct sun; reduce physical exertion.

In severe cases: cold dextrose saline IV, ice packs to ear margins.

7. Iron Deficiency Anaemia

Clinical Features & Diagnosis

Definition: Qualitative and quantitative reduction in circulating red blood cells and haemoglobin. In elephants, chronic dietary deficiency or parasitism are common underlying causes.

Finding	Significance
Pale mucous membranes	Reduced haemoglobin → less colour in conjunctiva, oral mucosa

Finding	Significance
Extreme weakness	Reduced oxygen-carrying capacity → poor cellular energy production
Inability to walk / slow movement	Muscle hypoxia; severe cases cannot rise
Blood smear — target cells	RBCs show central pallor with dense peripheral rim ("target cell" morphology)
Microcytic hypochromic RBCs	Small, pale cells on haematology — hallmark of iron-deficiency anaemia
Low haemoglobin (Hb)	Confirmed on complete blood count (CBC)

Treatment

- 1. Diet restoration** — full normal elephant ration: soyabeans, horse gram, napier grass, bamboo, leaves, grains, seasonal fruits.
- 2. Iron supplementation** — natural sources: jaggery (rich in iron), honey; plus oral or parenteral iron preparations as indicated.
- 3. Vitamin B12 injections** — essential cofactor for RBC maturation.
- 4. Multivitamin and mineral mixture** supplementation to correct concurrent micro-nutrient gaps.

8. Quick Reference — Conditions Covered

Condition	Category	Key Agent / Trigger	Hallmark Finding
Kodo millet poisoning	Toxic / mycotoxin	CPA from <i>Aspergillus/Penicillium</i>	Liver/kidney necrosis, cardiac failure, rapid death
HCN poisoning	Toxic / plant	Cyanogenic glycosides in sorghum/jowar	Restlessness, diarrhoea, respiratory distress
Cyanotoxins	Toxic / environmental	Cyanobacterial blooms in stagnant water	Neuro- and hepatotoxicosis; mass die-offs
Plant toxicity	Toxic / plant	Oleander (cardiac), calcium oxalate plants	Cardiac arrhythmia, neurological signs, GI irritation
Electrocution	Physical	Power-line / electrified-fence contact	Cardiac arrest, burns; rapid decomposition post-mortem
Pyometra	Reproductive	Hormonal + bacterial (uterine)	Uterine pus, systemic sepsis; older captive females
Hyperthermia	Thermal / metabolic	High temperature + no sweat glands	Collapse, organ failure; respond to cooling measures
Iron deficiency anaemia	Nutritional	Poor diet, parasitism, chronic disease	Pale mucosae, weakness, microcytic hypochromic RBCs

Key Takeaways

Chapter Summary — Non-Infectious Pathology in Asian Elephants

1. Elephant poisoning is both intentional (retaliation/poaching) and accidental (mycotoxins, toxic plants, algal blooms) — field history is critical to distinguish.
2. CPA from kodo millet is the most documented mycotoxin killing wild elephants in India; heavy rainfall + unharvested grain = highest risk.
3. HCN from jowar is treatable with sodium thiosulphate — suspect when standard anti-diarrhoeal therapy fails after recent sorghum feeding.
4. Electrocutation means death; distinguish from non-fatal electric shock at necropsy (burn marks, rapid decomposition, absent rigor mortis).
5. Pyometra targets aging, nulliparous captive females — hormonal + antibiotic treatment; surgical vestibulotomy in severe cases.
6. Hyperthermia is unique to large-bodied, sweat-gland-deficient animals; first response is water, shade, and active skin cooling.
7. Iron deficiency anaemia: confirm with blood smear (target cells, microcytic hypochromic RBCs); treat with diet correction + iron + B12.

CHAPTER 7

Postmortem Examination of the Asian Elephant

Protocol, Equipment, Systematic Examination & Differential Diagnosis

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Workshop on Essentials for Mortality Investigation of Asian Elephant · Chhattisgarh Forest Department · June 2026

1. Wildlife Health Management — Context

Wildlife health is an important but still-developing discipline that is often overlooked, even though it is a key component of wildlife conservation.

Definition

Definition (Shrivastav, 2010): Wildlife Health Management is a branch of veterinary science dealing with veterinary interventions in wild fauna — health monitoring, disease management, scientific immobilization, and habitat manipulation — to achieve conservation strategies that maintain a healthy population or conserve a particular species.

2. Introduction to Postmortem Examination

Terminology

Term	Definition
Necropsy	Postmortem examination of animals and birds.
Autopsy	Postmortem examination of humans.

Key Principles

- PM examination must be conducted as per a **standard protocol** and with **adequate facilities**.
- The carcass of a wild animal is **seldom found in a condition suitable** for PM examination — early detection and prompt response are critical.
- **Who should perform it?** An experienced wildlife veterinarian, or a veterinary pathologist with exposure to wildlife.
- Postmortem examination is an **art of disease diagnosis** — a scientific, systematic requirement for recording gross pathological changes in the carcass.
- **Gross lesions alone are not sufficient** for a definitive diagnosis. Laboratory support is essential: correct samples in appropriate preservatives.
- A badly putrefied sample — or even a fresh sample in an unsuitable preservative — will not be useful.

“Doctors who perform necropsies, or who regularly witness postmortem examinations, learn to solve their doubts — unlike those not dealing with necropsy, whose answers are left floating in the air.”

— Dr. A. B. Shrivastav, 2014

“Carcass never tells a lie.”

— Dr. Satpati

Staff Required

Role	Number
Wildlife veterinarians	2–3
Supporting staff	4
Photographer	1
Sample collection specialist	1

3. Necropsy Objectives

1. Identify all possible **lesions of disease or abnormality**.
2. Correlate lesions with **clinical and laboratory findings**.
3. Understand the **sequence of disease events** (pathogenesis).
4. Provide information to **prevent or control the disease** in the healthy population.

4. Pre-Necropsy Evaluation

Before beginning, examine the **necropsy request letter** for:

- Specific lesions desired in particular organs or systems.
- Whether a **cosmetic** (for display) or **routine** necropsy is required.
- Special requests — sample collection for specific investigations, cultures, photographs, radiographs, etc.
- **Identify the carcass** positively before beginning — confirm the correct individual is being necropsied.
- Review all available **clinical and circumstantial history**.

PM examinations of Schedule I animals must be performed systematically, sincerely, and seriously — never as a routine chore.

5. Protocol for PM Examination — General Workflow

Step	Action	Notes
1	Letter from the competent authority	Legal authorisation before any PM is commenced
2	Facilities	Appropriate site — good daylight, space for large carcass, drainage
3	Measurements	Body length, girth, tusk length; use calibrated instruments
4	Observation — External then Internal	Systematic external examination before any incision; then internal organ systems
5	Gross interpretation	Record and interpret all gross lesions before collecting samples

Step	Action	Notes
6	Sample collection	Collect biological samples in correct preservatives immediately after gross review
7	Laboratory submission	Suitable samples dispatched with completed history form and transport certificate
8	Report writing	Tentative cause of death → confirmatory lab results → final determination

General Logistics — Non-Negotiable Requirements

Always perform in good daylight.

Use standard large-animal necropsy instruments (stainless steel, sharp).

Protective clothing (PPE) mandatory for all veterinary and support staff.

Maintain photographic and video documentation throughout.

6. Equipment Required

Documentation & Measurement Items

Item	Min. Qty
Copy of necropsy protocol	1
Hand-held GPS	1
Digital camera	1
Digital video camera	1
Clipboard	2
Measuring tape	1
Plastic ruler 15 cm (clear markings)	2
Vernier calipers	1
Portable weighing machine (250 kg)	1
Necropsy format (data sheets)	5
Ballpen	2
Pencil + rubber	1 set
Metal detector	1

Instruments (Stainless Steel — Best Quality)

Instrument	Min. Qty
Sharp high-quality necropsy knife (SS)	4
Skinning knife, curved (SS)	2
Autopsy knife, curved (SS)	2
Slicing knife, 22 mm blade (SS)	1
Knife sharpener stone or steel	1

Instrument	Min. Qty
Bard-Parker handle + blades (No. 20, 22, 26)	5 each
Small plain forceps	As needed
Artery forceps, straight 203 mm (SS)	2
Dissecting forceps, 200 mm (SS)	2
Large Mayo dissecting scissors, fine point 215 mm	2
Bone cutter, compound action 266 mm	1
Hack saw / bone saw blade 254 mm	1
Chisel 22 cm × 3 cm blade	1
Hammer, wrench end 200 mm	1
Portable autopsy saw (electric/battery)	1
Axe (roofing axe, SS) 60 mm blade	1
Spirit lamp (SS) or gas burner	1

7. External Examination

General Assessment

- **Body condition score** — assess fat cover, muscle mass.
- **Rigor mortis** — presence, degree, and stage.
- **External injuries** — characterize location, type, age of injury.
- **Documentation** — photograph every finding before disturbing.
- **Any other observation** relevant to circumstances of death.

Detailed Checklist (11 Points)

Check	What to Assess
1. Sex identification	Examine male/female external organs; note pregnancy, lactation, or dry status.
2. Nutritional status	Prominent ribs, depression in lumbar and buccal regions, distinct temporal fossa, and loose skin all indicate poor nutritional status.
3. Fractures & dislocations	Search for fractures and dislocations systematically where possible.
4. Postmortem changes	Assess algor mortis (cooling), rigor mortis (stiffening), livor mortis (gravitational blood pooling), and autolytic changes.
5. Mucous membranes	Examine for pallor (anaemia), congestion (hyperaemia), cyanosis (hypoxia), and inflammatory conditions.
6. Skin	Check for external parasites, cutaneous filariasis, pox lesions, oedematous swellings, emphysematous crepitation, injuries, burns, and warts.
7. Injuries	Look for bullet wounds, intraspecies fight injuries, large-carnivore mauling (especially in calves), and ankush (hook) injuries in captive elephants.
8. Dung bolus at rectum	Indicates digestive function, dentition status, and hydration — dry, hard pellets vs. loose, watery faeces are both informative.

Check	What to Assess
9. Natural orifices	Check all orifices for discharges. Examine oral cavity for FMD lesions, EEHV ulcers, anthrax signs, etc.
10. Temporal glands	Examine for musth secretions in adult males — relevant to circumstances of death (fighting, restraint-related stress).
11. Foot pads and nails	Check for injuries, cracks, foreign bodies, and overall condition.

8. Postmortem Changes

Change	Description	Timeline / Notes
Cadaveric lividity (Hypostatic congestion)	Irregular patches of dark discolouration in subcutaneous tissues on the dependent (lower) side — particularly prominent in large animals like elephants due to large blood volume. Must be distinguished from true haemorrhages.	Begins within 1–2 hours; fixed by 6–8 hours (cannot be moved by repositioning).
Rigor mortis	Progressive stiffening of muscle due to ATP depletion and actin-myosin crosslinking. Less marked in weak, emaciated, diseased, or septicemic animals.	Appears 1–4 hours post-death; lasts 16–24 hours; may persist up to 48 hours depending on temperature and nutritional state.
Autolytic changes	Cellular self-digestion by endogenous enzymes released from disrupted lysosomes. Rate depends on atmospheric moisture, temperature, and postmortem interval.	Begins immediately after death; visible as green discolouration, gas bloat, softening of tissues, and odour.

9. Dentition & Age Determination

Elephants develop **six successive sets of molars (M1–M6)** that erupt and wear sequentially throughout life. Each set replaces the previous as it wears down — unlike most mammals where molars erupt once. The number of **laminae** (enamel ridges) increases with each molar and can be used to estimate age.

Molar	Approx. Age at Eruption	Laminae (Enamel Ridges)
M1	~1 year	5
M2	~2 years	7
M3	~6 years	10
M4	~15 years	10
M5	~28 years	12
M6	~47 years	13

Tusk & Final Molar Note

Permanent tusks protrude beyond the lips at approximately 30 months and continue growing throughout life — tusk length can supplement age estimation. Once M6 wears out (typically 60+ years), the elephant can no longer chew and eventually dies of malnutrition.

10. Skinning the Carcass

In a healthy animal the skin is thick but flexible and moves easily over the underlying tissues. Proper skinning is essential before internal examination.

- Remove the skin of **one side** — from the entire lateral thoracic region down to the lower abdominal and other suspected regions.
- **Examine subcutaneous tissues and superficial muscles** for abscesses, wounds, haemorrhages, and other pathological conditions.
- Identify and examine all **superficial lymph nodes** before proceeding to the thoracic and abdominal cavities.

11. Internal Examination — Systematic Approach

System	Key Organs & Examination Points
Respiratory	Trachea — examine for mucus, haemorrhage, foreign body. Lungs — consistency, colour, pleural surface, cut surface (oedema, consolidation, haemorrhage).
Digestive	Pharynx, oesophagus, stomach, small intestine, large intestine, caecum, rectum, anus. Also: molar teeth and tongue, salivary glands, liver, pancreas. Note: the elephant is a simple-stomached (monogastric) animal. Stomach length $\approx$ 100–140 cm; diameter $\approx$ 40 cm. Examine stomach contents — food material, moisture, parasites, ulcers, gastritis.
Urogenital	Kidneys — size, colour, capsule, cut surface (cortex-medulla ratio). Bladder — distension, mucosa, content.
Reproductive	Uterus — size, content, endometrium. Ovaries / testes — as appropriate.
Musculoskeletal	Muscles — pallor, haemorrhage, necrosis, parasites. Bones — fractures, metabolic changes.
Nervous	Brain — meninges, congestion, herniation, grey/white matter changes.
Cardiovascular	Heart — size, epicardium, myocardium (cut surface), valves, great vessels.

Anatomical Note — Elephant Abdomen

Abdominal cavity approach: draw a vertical line from the last rib to the lower abdomen; make a vertical incision down to ground surface to open the cavity.
The elephant has TWENTY PAIRS of ribs (vs. 13 in most large mammals).

12. Lymph Node Examination

Superficial lymph nodes are particularly important for detecting infectious diseases such as **tuberculosis** and **EEHV**. Thick elephant skin makes external palpation difficult; systematic incision is required.

Superficial Lymph Nodes

Node	Location	Notes
Mandibular	At the ventral border of the lower jaw, just under the skin.	Incise between the mandibles and pull the tongue forward during oral cavity inspection.
Parotid	Just below the ear, near the parotid salivary gland.	Enlarged in head and neck infections.

Node	Location	Notes
Superficial cervical	On the lateral side of the neck, slightly cranial to the shoulder.	Drains superficial neck structures.
Prescapular	Just ahead of the shoulder joint.	Drains the forelimb and cranial thorax.
Femoral	Close to the head of the femur.	Drains the hindlimb.

Deep / Internal Lymph Nodes

Node	Location	Clinical Significance
Bronchial (Tracheobronchial)	At the carina — where the bronchi bifurcate from the trachea.	MUST be sampled for mycobacterial culture (TB). First site of lung lymph drainage; enlarged and caseous in TB.
Hepatic & splenic	Near the liver (hepatic hilus) and spleen respectively.	Checked for systemic infection, haemorrhage, and reactive hyperplasia.
Mesenteric	Clusters distributed in the mesentery throughout the bowel.	Filter lymph from the intestines; trap bacteria, viruses, and antigens before they enter the bloodstream. Key in enteric infections.

13. Histopathology Highlight — EEHV

Elephant Endotheliotropic Herpesvirus (EEHV) is one of the most dangerous pathogens of young Asian elephants. Histopathological examination of affected endothelial tissues — particularly from the tongue, heart, and trunk tip — may reveal **intranuclear inclusion bodies**, the hallmark of herpesvirus infection.

EEHV — Histopathology & Diagnosis

Intranuclear inclusion bodies in vascular endothelium = EEHV until proven otherwise.

Collect trunk wash, oral and rectal swabs, and fresh endothelial-rich tissue (tongue tip, trunk tip) for PCR confirmation.

A T-cell-inducing heterologous vaccine against EEHV is currently under trial study.

14. Differential Diagnosis — Sudden Onset, Rapid Death

Disease	Characteristic Finding	Key Samples
Haemorrhagic septicaemia (Pasteurellosis)	Extensive haemorrhagic areas throughout the carcass; acute gelatinous oedema in subcutaneous tissues of the neck and brisket.	Swabs, blood smear, culture from lymph node and lung.
Salmonellosis	Watery diarrhoea with or without blood; small intestinal mucosal haemorrhage and necrosis; mesenteric lymph node enlargement.	Intestinal loop culture; fresh faeces; mesenteric LN.
Enterotoxaemia (Clostridial)	Profuse diarrhoea leading to rapid dehydration and death; bloat; haemorrhagic intestinal contents.	Intestinal loop in formalin; fresh intestinal content for toxin assay.
Encephalomyocarditis (EMCV)	Cardiac muscle pallor and necrosis; respiratory signs may be observed; sudden death in young/stressed elephants.	Heart tissue (fresh and fixed); serum for virus isolation/PCR.

15. Case Example — Rabies in Semi-Captive Elephant (Panna TR)

A semi-captive elephant at Panna Tiger Reserve presented with the following sequence of clinical signs:

Time Point	Clinical Event
Initially	Off feed, restlessness, excessive drooling of saliva.
Day 3	Restricted jaw movement, protrusion of tongue.
Day 4	Hyperaemic conjunctiva (intense reddening of the eyes).
Day 5	Hydrophobia; loud but markedly changed/abnormal trumpeting.
Day 6	Lateral recumbency and death.
Post-death	Proper carcass disposal under biosafety protocols.

Rabies — Sample & Biosafety Notes

Rabies in elephants is rare but documented. Submit fresh brain tissue in two portions: (1) half in 10% NBF for histopathology (Negri body detection), (2) half fresh/frozen for fluorescent antibody test (FAT) or RT-PCR. Handle with full Level-3 PPE. Carcass disposal must follow zoonotic disease biosafety protocols.

16. Other Conditions of Note

Condition	Notes for PM Examination
Tuberculosis (TB)	Gross lesions — caseous nodules in lung and tracheobronchial lymph nodes — documented in free-ranging Asian elephants. Always sample bronchial lymph nodes for mycobacterial culture.
HCN (hydrocyanic acid) poisoning	Common in herbivores fed sorghum/jowar species; cyanogenic glycoside content highest in young, rapidly growing leaves and stalks (after autumn rains or crop regrowth). Managed with sodium thiosulphate + supportive therapy. (Ref: Shrivastav, Zoo Print, Vol. XXV, June 2010.)
Decomposed / waterlogged carcass	Severely autolysed or submerged carcasses may make assessment "difficult to advise" — tissue architecture is destroyed, cultures contaminated, and toxicological results unreliable. Emphasises the value of early detection.

Key Takeaways

Chapter Summary — Postmortem Examination of the Asian Elephant

1. A wildlife necropsy is only as good as its protocol, facilities, and operator experience. Always work in good daylight with complete PPE.
2. Gross findings alone rarely yield a definitive diagnosis — confirmatory diagnosis depends on correctly collected and preserved laboratory samples.
3. For elephants: use molars/tusks for age estimation; remember the monogastric stomach (100–140 cm); and always note the 20 pairs of ribs.
4. Prioritise tracheobronchial lymph node sampling for TB and trunk wash / endothelial tissue for EEHV PCR confirmation.
5. Cadaveric lividity must be distinguished from true haemorrhagic lesions.
6. Rabies carries zoonotic risk — handle with Level-3 PPE; dispose of carcass under biosafety protocols.
7. "The carcass never tells a lie" — but only if the necropsy is performed promptly, systematically, and with full documentation.